43

Molecules and Solids

CHAPTER OUTLINE

43.1 Molecular Bonds

43.2 Energy States and Spectra of Molecules

43.3 Bonding in Solids

43.4 Free-Electron Theory of Metals

43.5 Band Theory of Solids

43.6 Electrical Conduction in Metals, Insulators, and Semiconductors

43.7 Semiconductor Devices

43.8 Superconductivity

* An asterisk indicates a question or problem new to this edition.

ANSWERS TO OBJECTIVE QUESTIONS

OQ43.1 (a) False. An infinite current would produce an infinite magnetic field that would penetrate the surface of the superconductor and destroy the superconducting properties. (b) False. There is no physical requirement that a superconductor carry a current. (c) True. (d) True. (e) True. Collisions do not occur between Cooper pairs and the lattice ions.

OQ43.2 Answer (b). At higher temperature, molecules are typically in higher rotational energy levels before as well as after infrared absorption.

OQ43.3 **(i)** Answer (c). Think of aluminum foil.

(ii) Answer (a). An example is NaCl, table salt.

(iii) Answer (b). Examples are elemental silicon and carborundum (silicon carbide).

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OQ43.4 **(i)** Answer (b). The density of states is proportional to the energy to the one-half power.

(ii) Answer (a). Most states well above the Fermi energy are unoccupied.

OQ43.5 Answer (b). First consider electric conduction in a metal. The number of conduction electrons is essentially fixed. They conduct electricity by having drift motion in an applied electric field superposed on their random thermal motion. At higher temperature, the ion cores vibrate more and scatter more efficiently the conduction electrons flying among them. The mean time between collisions is reduced. The electrons have time to develop only a lower drift speed. The electric current is reduced, so we see the resistivity increasing with temperature.

Now consider an intrinsic semiconductor. At absolute zero its valence band is full and its conduction band is empty. It is an insulator, with very high resistivity. As the temperature increases, more electrons are promoted to the conduction band, leaving holes in the valence band. Then both electrons and holes move in response to an applied electric field. Thus we see the resistivity decreasing as temperature goes up.

OQ43.6 **(i)** and **(ii)** Answer (a) for both. Either kind of doping contributes more mobile charge carriers, either holes or electrons.

OQ43.7 The ranking is then b > d > c > a. If you start with a solid sample and raise its temperature, it will typically melt first, then start emitting lots of far infrared light, then emit light with a spectrum peaking in the near infrared, and later have its molecules dissociate into atoms. Rotation of a diatomic molecule involves less energy than vibration. Absorption and emission of microwave photons, of frequency ~10^{11} Hz, accompany excitation and de-excitation of rotational motion, while infrared photons, of frequency ~10^{13} Hz, accompany changes in the vibration state of typical simple molecules.

ANSWERS TO CONCEPTUAL QUESTIONS

CQ43.1 A material can absorb a photon of energy greater than the energy gap, as an electron jumps into a higher energy state; therefore, silicon can absorb visible light, thus appearing opaque. If the photon does not have enough energy to raise the energy of the electron by the energy gap, then the photon will not be absorbed; therefore, diamond cannot absorb visible light, thus appearing transparent.

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CQ43.2 Rotational, vibrational, and electronic (as discussed in Chapter 42) are the three major forms of excitation. Rotational energy for a diatomic molecule is on the order of $\frac{\hbar^2}{2I}$, where I is the moment of inertia of the molecule. A typical value for a small molecule is on the order of 1 meV = 10^{-3} eV. Vibrational energy is on the order of hf, where f is the vibration frequency of the molecule. A typical value is on the order of 0.1 eV. Electronic energy depends on the state of an electron in the molecule and is on the order of a few eV. The rotational energy can be zero, but neither the vibrational nor the electronic energy can be zero.

CQ43.3 From the rotational spectrum of a molecule, one can easily calculate the moment of inertia of the molecule using Equation 43.7 in the text. Note that with this method, only the spacing between adjacent energy levels needs to be measured. From the moment of inertia, the size of the molecule can be calculated, provided that the structure of the molecule is known.

CQ43.4 Along with arsenic (As), any other element in group V, such as phosphorus (P), antimony (Sb), and bismuth (Bi), would make good donor atoms. Each has 5 valence electrons. Any element in group III would make good acceptor atoms, such as boron (B), aluminum (Al), gallium (Ga), and indium (In). They all have only 3 valence electrons.

CQ43.5 The energy of the photon is given to the electron. The energy of a photon of visible light is sufficient to promote the electron from the lower-energy valence band to the higher-energy conduction band. This results in the additional electron in the conduction band and an additional hole—the energy state that the electron used to occupy—in the valence band.

CQ43.6 (a) In a metal, there is no energy gap between the valence and conduction bands, or the conduction band is partly full even at absolute zero in temperature. Thus an applied electric field is able to inject a tiny bit of energy into an electron to promote it to a state in which it is moving through the metal as part of an electric current. In an insulator, there is a large energy gap between a full valence band and an empty conduction band. An applied electric field is unable to give electrons in the valence band enough energy to jump across the gap into the higher energy conduction band. In a semiconductor, the energy gap between valence and conduction bands is smaller than in an insulator.

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(b) At absolute zero the valence band is full and the conduction band is empty, but at room temperature thermal energy has promoted some electrons across the gap. Then there are some mobile holes in the valence band as well as some mobile electrons in the conduction band.

CQ43.7 (a) The two assumptions in the free-electron theory are that the conduction electrons are not bound to any particular atom, and that the nuclei of the atoms are fixed in a lattice structure. In this model, it is the "soup" of free electrons that are conducted through metals.

(b) The energy band model is more comprehensive than the free-electron theory. The energy band model includes an account of the more tightly bound electrons as well as the conduction electrons. It can be developed into a theory of the structure of the crystal and its mechanical and thermal properties.

CQ43.8 A molecule containing two atoms of D = ^{2}H, deuterium, has twice the mass of a molecule containing two atoms of ordinary hydrogen ^{1}H; therefore the deuterium molecule has twice the reduced mass of the hydrogen molecule. The atoms have the same electronic structure, so the molecules have the same interatomic spacing, and the same spring constant. Therefore, each vibrational energy level for D_2 is $1/\sqrt{2}$ times that of H_2. The moment of inertia of deuterium is twice as large and the rotational energies one-half as large as for the ordinary hydrogen molecule.

CQ43.9 Ionic bonds are ones between oppositely charged ions. One atom essentially steals an electron from another; for example, in table salt, NaCl, the chlorine atom takes the outer 3*s* electron from the sodium atom, resulting in two ions Cl^- and Na^+. A simple model of an ionic bond is the electrostatic attraction of a negatively charged latex balloon to a positively charged Mylar balloon.

Covalent bonds are ones in which atoms share electrons. Classically, two children playing a short-range game of catch with a ball models a covalent bond. On a quantum scale, the two atoms are sharing a wave function, so perhaps a better model would be two children using a single hula hoop.

Van der Waals bonds are weak electrostatic forces: the electric dipole-dipole force is analogous to the attraction between the opposite poles of two bar magnets, the dipole-induced dipole force is similar to a bar magnet attracting an iron nail or paper clip, and the dispersion force is analogous to an alternating-current electromagnet attracting a paper clip.

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A hydrogen atom in a molecule is not ionized, but its electron can spend more time elsewhere than it does in the hydrogen atom. The hydrogen atom can be a location of net positive charge, and can weakly attract a zone of negative charge in another molecule.

CQ43.10 The atoms of crystalline substances form a regular array of ions in a lattice structure, and the atoms are close enough together to allow energy bands to form. The atoms of amorphous solids do not form a regular array, but they are close enough to produce energy bands. The atoms of gases do not form regular arrays and are too far apart to form energy bands.

SOLUTIONS TO END-OF-CHAPTER PROBLEMS

Section 43.1 Molecular Bonds

P43.1 At the boiling or condensation temperature,

$$E = \frac{3}{2}k_B T \approx 10^{-3}\ \text{eV} = 10^{-3}\left(1.6\times10^{-19}\ \text{J}\right)$$

Solving for the temperature T gives,

$$T = \frac{E}{k_B} \approx \frac{2\left(1.6\times10^{-22}\ \text{J}\right)}{3\left(1.38\times10^{-23}\ \text{J/K}\right)} \boxed{\sim 10\ \text{K}}$$

P43.2 (a) The electrostatic force is

$$F = \frac{q^2}{4\pi\epsilon_0 r^2}$$

$$= \frac{\left(8.99\times10^{9}\ \text{N}\cdot\text{m}^2/\text{C}^2\right)\left(1.60\times10^{-19}\ \text{C}\right)^2}{\left(5.00\times10^{-10}\ \text{m}\right)^2}$$

$$= 9.21\times10^{-10}\ \text{N} = 921\times10^{-12}\ \text{N}$$

or $\boxed{921\ \text{pN toward the other ion.}}$

(b) The potential energy of the ion pair is

$$U = \frac{-q^2}{4\pi\epsilon_0 r}$$

$$= -\left[\frac{\left(8.99\times10^{9}\ \text{N}\cdot\text{m}^2/\text{C}^2\right)\left(1.60\times10^{-19}\ \text{C}\right)^2}{5.00\times10^{-10}\ \text{m}}\right]\left(\frac{1\ \text{eV}}{1.60\times10^{-19}\ \text{J}}\right)$$

$$= \boxed{-2.88\ \text{eV}}$$

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P43.3 We are told that

$$K + Cl + 0.70\ eV \rightarrow K^+ + Cl^-$$

and $$Cl + e^- \rightarrow Cl^- + 3.6\ eV$$

or $$Cl^- \rightarrow Cl + e^- - 3.6\ eV$$

By substitution,

$$K + Cl + 0.7\ eV \rightarrow K^+ + Cl + e^- - 3.6\ eV$$

$$K + 4.3\ eV \rightarrow K^+ + e^-$$

or the ionization energy of potassium is $\boxed{4.3\ eV}$.

P43.4 (a) Because the ionization energy of K is 4.34 eV, we have the relation

$$K + 4.34\ eV \rightarrow K^+ + e^- \qquad \textbf{[1]}$$

and because the electron affinity of I is 3.06 eV, we have the relation

$$I + e^- \rightarrow I^- + 3.06\ eV$$
$$I - 3.06\ eV \rightarrow I^- - e^- \qquad \textbf{[2]}$$

Adding equations [1] and [2] gives

$$(K + 4.34\ eV) + (I - 3.06\ eV) \rightarrow (K^+ + e^-) + (I^- - e)$$

$$K + I + (4.34\ eV - 3.06\ eV) \rightarrow K^+ + I^-$$

$$K + I + 1.28\ eV \rightarrow K^+ + I^-$$

Therefore, the activation energy is $\boxed{E_a = 1.28\ eV}$.

(b) We differentiate the given function:

$$\frac{dU}{dr} = \frac{4\epsilon}{\sigma}\left[-12\left(\frac{\sigma}{r}\right)^{13} + 6\left(\frac{\sigma}{r}\right)^{7}\right]$$

Setting the expression above equal to 0, at $r = r_0$ we have

$$\frac{dU}{dr} = 0 \quad \rightarrow \quad \left(\frac{\sigma}{r_0}\right)^{13} = \frac{1}{2}\left(\frac{\sigma}{r_0}\right)^{7}$$

which gives

$$\left(\frac{\sigma}{r_0}\right)^{6} = 2^{-1} \quad \rightarrow \quad \sigma = 2^{-1/6} r_0 = 2^{-1/6}(0.305)\ \text{nm}$$

or $\boxed{\sigma = 0.272\ \text{nm}}$

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Then also

$$U(r_0) = 4\epsilon\left[\left(\frac{2^{-1/6}r_0}{r_0}\right)^{12} - \left(\frac{2^{-1/6}r_0}{r_0}\right)^{6}\right] + E_a$$

$$= 4\epsilon\left[\frac{1}{4} - \frac{1}{2}\right] + E_a = -\epsilon + E_a$$

solving for ϵ gives

$$\epsilon = E_a - U(r_0) = 1.28\text{ eV} + 3.37\text{ eV}$$

$$= \boxed{4.65\text{ eV}}$$

(c) The force of attraction between the atoms is

$$F(r) = -\frac{dU}{dr} = \frac{4\epsilon}{\sigma}\left[12\left(\frac{\sigma}{r}\right)^{13} - 6\left(\frac{\sigma}{r}\right)^{7}\right]$$

To find the maximum force we calculate

$$\frac{dF}{dr} = \frac{4\epsilon}{\sigma^2}\left[-156\left(\frac{\sigma}{r}\right)^{14} + 42\left(\frac{\sigma}{r}\right)^{8}\right] = 0$$

$$\frac{\sigma}{r_{\text{break}}} = \left(\frac{42}{156}\right)^{1/6}$$

So at $r = r_{\text{break}}$, the force is a maximum:

$$F_{\max} = \frac{4(4.65\text{ eV})}{0.272\text{ nm}}\left[12\left(\frac{42}{156}\right)^{13/6} - 6\left(\frac{42}{156}\right)^{7/6}\right]$$

$$= \frac{-41.0\text{ eV}}{\text{nm}}\left(\frac{1.60\times10^{-19}\text{ N}\cdot\text{m}}{1\text{ eV}}\right)\left(\frac{1\text{ nm}}{10^{-9}\text{ m}}\right) = -6.55\text{ nN}$$

Therefore the applied force required to break the molecule is $\boxed{+6.55\text{ nN}}$ away from the center.

(d) To calculate the force constant, we expand $U(r)$ as suggested in the problem statement:

$$U(r_0 + s) = 4\epsilon\left[\left(\frac{\sigma}{r_0 + s}\right)^{12} - \left(\frac{\sigma}{r_0 + s}\right)^{6}\right] + E_a$$

$$= 4\epsilon\left[\left(\frac{2^{-1/6}r_0}{r_0 + s}\right)^{12} - \left(\frac{2^{-1/6}}{r_0 + s}\right)^{6}\right] + E_a$$

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Expanding,

$$U(r_0+s)=4\epsilon\left[\frac{1}{4}\left(1+\frac{s}{r_0}\right)^{-12}-\frac{1}{2}\left(1+\frac{s}{r_0}\right)^{-6}\right]+E_a$$

$$=4\epsilon\left[\frac{1}{4}\left(1-12\frac{s}{r_0}+78\frac{s^2}{r_0^2}-\cdots\right)\right.$$

$$\left.-\frac{1}{2}\left(1-6\frac{s}{r_0}+21\frac{s^2}{r_0^2}-\cdots\right)\right]+E_a$$

$$=\ \epsilon-12\epsilon\frac{s}{r_0}+78\epsilon\frac{s^2}{r_0^2}-2\epsilon+12\epsilon\frac{s}{r_0}-42\epsilon\frac{s^2}{r_0^2}+E_a+\cdots$$

$$=-\epsilon+E_a+0\left(\frac{s}{r_0}\right)+36\epsilon\frac{s^2}{r_0^2}+\cdots$$

or $\quad U(r_0+s)\approx U(r_0)+\frac{1}{2}ks^2$

where $k=\dfrac{72\epsilon}{r_0^2}=\dfrac{72(4.65\text{ eV})}{(0.305\text{ nm})^2}=3\,599\text{ eV/nm}^2=\boxed{576\text{ N/m}}$

P43.5 (a) The minimum energy of the molecule at $r = r_0$ is found from

$$\frac{dU}{dr}=-12Ar_0^{-13}+6Br_0^{-7}=0$$

yielding

$$r_0=\left[\frac{2A}{B}\right]^{1/6}$$

$$=\left[\frac{2\left(0.124\times10^{-120}\text{ eV}\cdot\text{m}^{12}\right)}{1.488\times10^{-60}\text{ eV}\cdot\text{m}^6}\right]^{1/6}=7.42\times10^{-11}\text{ m}=\boxed{74.2\text{ pm}}$$

(b) The energy required to break up the molecule would separate the atoms from $r = r_0$ to $r = \infty$:

$$E=U|_{r=\infty}-U|_{r=r_0}=0-\left[\frac{A}{4A^2/B^2}-\frac{B}{2A/B}\right]=-\left[\frac{1}{4}-\frac{1}{2}\right]\frac{B^2}{A}=\frac{B^2}{4A}$$

$$E=\frac{\left(1.488\times10^{-60}\text{ eV}\cdot\text{m}^6\right)^2}{4\left(0.124\times10^{-120}\text{ eV}\cdot\text{m}^{12}\right)}=\boxed{4.46\text{ eV}}$$

This is also the equal to the binding energy, the amount of energy given up by the two atoms as they come together to form a molecule.

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P43.6 (a) The minimum energy of the molecule at $r = r_0$ is found from

$$\frac{dU}{dr} = -12Ar_0^{-13} + 6Br_0^{-7} = 0$$

yielding

$$r_0 = \left[\frac{2A}{B}\right]^{1/6}$$

(b) The energy required to break up the molecule would separate the atoms from $r = r_0$ to $r = \infty$:

$$E = U|_{r=\infty} - U|_{r=r_0} = 0 - \left[\frac{A}{4A^2/B^2} - \frac{B}{2A/B}\right] = -\left[\frac{1}{4} - \frac{1}{2}\right]\frac{B^2}{A} = \boxed{\frac{B^2}{4A}}$$

Section 43.2 Energy States and Spectra of Molecules

P43.7 (a) Recall from Chapter 42 that the energy of the photon is given by

$$hf = \Delta E = \frac{\hbar^2}{2I}\left[2(2+1)\right] - \frac{\hbar^2}{2I}\left[1(1+1)\right] = \frac{\hbar^2}{2I}(4)$$

Then,

$$I = \frac{4\left(h/2\pi\right)^2}{2hf} = \frac{h}{2\pi^2 f} = \frac{6.626\times10^{-34}\ \text{J}\cdot\text{s}}{2\pi^2\left(2.30\times10^{11}\ \text{Hz}\right)}$$

$$= \boxed{1.46\times10^{-46}\ \text{kg}\cdot\text{m}^2}$$

(b) The results are the same, suggesting that the bond length of the molecule does not change measurably between the two transitions.

P43.8 From Equations 43.4 and 43.3, the reduced mass and moment of inertia of CsI are

$$\mu = \frac{m_1 m_2}{m_1 + m_2} = \frac{(132.9\ \text{u})(126.9\ \text{u})}{132.9\ \text{u} + 126.9\ \text{u}}\left(\frac{1.66\times10^{-27}\ \text{kg}}{\text{u}}\right)$$

$$= 1.08\times10^{-25}\ \text{kg}$$

and

$$I = \mu r^2 = \left(1.08\times10^{-25}\ \text{kg}\right)\left(0.127\times10^{-9}\ \text{m}\right)^2$$

$$= 1.74\times10^{-45}\ \text{kg}\cdot\text{m}^2$$

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The allowed rotational energies (from Equation 43.6) are

$$E_{\text{rot}} = \frac{\hbar^2}{2I}J(J+1) = J(J+1)\left[\frac{\left(6.626\times10^{-34}\ \text{J}\cdot\text{s}/2\pi\right)^2}{2\left(1.74\times10^{-45}\ \text{kg}\cdot\text{m}^2\right)}\right]$$

$$= J(J+1)\left(3.20\times10^{-24}\ \text{J}\right)\cdot\left(\frac{1\ \text{eV}}{1.602\times10^{-19}\ \text{J}}\right)$$

$$= J(J+1)\left(2.00\times10^{-5}\ \text{eV}\right)$$

(a) $J = 2$ gives

$$E_{\text{rot}} = 2(3)\frac{\hbar^2}{2I} = 1.20\times10^{-4}\ \text{eV} = \boxed{0.120\ \text{meV}}$$

(b) The photon that can cause the transition $J = 1 \rightarrow 2$ has energy

$$hf = \Delta E_{\text{rot}} = 2(2+1)\left(\frac{\hbar^2}{2I}\right) - 1(1+1)\left(\frac{\hbar^2}{2I}\right) = 4\left(\frac{\hbar^2}{2I}\right)$$

$$= 4\left(3.20\times10^{-24}\ \text{J}\right) = 1.28\times10^{-23}\ \text{J} = 7.99\times10^{-2}\ \text{eV}$$

The frequency of the photon is

$$f = \frac{\Delta E_{\text{rot}}}{h} = \frac{1.28\times10^{-23}\ \text{J}}{6.626\times10^{-34}\ \text{J}\cdot\text{s}} = 1.93\times10^{10}\ \text{s}^{-1} = \boxed{19.3\ \text{GHz}}$$

***P43.9** For the HCl molecule in the $J = 2$ rotational energy level, we are given the distance between nuclei, $r_0 = 0.127\ 5$ nm. From Equation 43.6, the allowed rotational energies are

$$E_{\text{rot}} = \frac{\hbar^2}{2I}J(J+1)$$

Taking $J = 2$, we have $E_{\text{rot}} = 6\frac{\hbar^2}{2I} = \frac{3\hbar^2}{I} = \frac{1}{2}I\omega^2$,

or $$\omega = \sqrt{\frac{6\hbar^2}{I^2}} = \sqrt{6}\,\frac{\hbar}{I}$$

The moment of inertia of the molecule is given by Equation 43.3:

$$I = \mu r_0^2 = \left(\frac{m_1 m_2}{m_1 + m_2}\right)r_0^2$$

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Substituting numerical values,

$$I = \left[\frac{(1.008\ \text{u})(35.45\ \text{u})}{1.008\ \text{u} + 35.45\ \text{u}}\right] r_0^2$$
$$= (0.980\ \text{u})\left(1.66 \times 10^{-27}\ \text{kg/u}\right)\left(1.275 \times 10^{-10}\ \text{m}\right)^2$$
$$= 2.64 \times 10^{-47}\ \text{kg} \cdot \text{m}^2$$

Therefore,

$$\omega = \sqrt{6}\frac{\hbar}{I} = \frac{\sqrt{6}\left(6.626 \times 10^{-34}\ \text{J} \cdot \text{s}\right)}{2\pi(2.64 \times 10^{-47}\ \text{kg} \cdot \text{m}^2)} = \boxed{9.77 \times 10^{12}\ \text{rad/s}}$$

P43.10 (a) From Equation 43.10, the energy separation between the ground and first excited state is

$$\Delta E_{\text{vib}} = \frac{h}{2\pi}\sqrt{\frac{k}{\mu}} = hf \qquad \text{so} \qquad k = 4\pi^2 f^2 \mu$$

and the reduced mass is

$$\mu = \frac{k}{4\pi^2 f^2} = \frac{1530\ \text{N/m}}{4\pi^2(56.3 \times 10^{12}\ \text{s}^{-1})^2} = \boxed{1.22 \times 10^{-26}\text{kg}}$$

(b) From Equation 43.4, the reduced mass is

$$\mu = \frac{m_1 m_2}{m_1 + m_2} = \frac{(14.007\ \text{u})(15.999\ \text{u})}{14.007\ \text{u} + 15.999\ \text{u}}\left(\frac{1.66 \times 10^{-27}\ \text{kg}}{1\ \text{u}}\right)$$
$$= \boxed{1.24 \times 10^{-26}\ \text{kg}}$$

(c) They agree because the small apparent difference can be attributed to uncertainty in the data.

P43.11 (a) With r representing the distance of each atom from the center of mass, the moment of inertia is

$$mr^2 + mr^2 = 2mr^2$$
$$= 2(1.008\ \text{u})\left(\frac{1.66 \times 10^{-27}\ \text{kg}}{\text{u}}\right)\left(\frac{0.750 \times 10^{-10}\ \text{m}}{2}\right)^2$$
$$= 4.71 \times 10^{-48}\ \text{kg} \cdot \text{m}^2$$

The allowed rotational energies (from Equation 43.6) are

$$E_{\text{rot}} = \frac{\hbar^2}{2I}J(J+1)$$

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The $J = 0$ state has energy $E_{\text{rot}} = 0$, and the $J = 1$ state has energy

$$E_{\text{rot}} = \frac{(h/2\pi)^2}{2I}(1)(2) = \frac{(h/2\pi)^2}{I}$$
$$= \frac{(6.626\times10^{-34}\ \text{J}\cdot\text{s}/2\pi)^2}{(4.71\times10^{-48}\ \text{kg}\cdot\text{m}^2)}\left(\frac{1\ \text{eV}}{1.602\times10^{-19}\ \text{J}}\right)$$
$$= 1.48\times10^{-2}\ \text{eV} = \boxed{0.014\ 8\ \text{eV}}$$

(b) The energy of the photon that raises the molecule from 0 to 0.014 8 eV is 0.014 8 eV. The photon's wavelength is

$$\lambda = \frac{h}{E} = \frac{1240\ \text{eV}\cdot\text{nm}}{0.0148\ \text{eV}} = 8.41\times10^4\ \text{nm}$$
$$= 84.1\times10^3\ \text{nm} = \boxed{84.1\ \mu\text{m}}$$

***P43.12** From Equation 43.10, the energy separation between the ground and first excited state is

$$\Delta E_{\text{vib}} = \hbar\sqrt{\frac{k}{\mu}} = \hbar\sqrt{\frac{k(m_1 + m_2)}{m_1 m_2}}$$

Substituting numerical values,

$$\Delta E_{\text{vib}} = (1.055\times10^{-34}\ \text{J}\cdot\text{s})\sqrt{\frac{(480\ \text{N/m})(35+1)}{(35)(1)(1.66\times10^{-27}\ \text{kg})}}$$
$$= 5.75\times10^{-20}\ \text{J} = 0.359\ \text{eV}$$

To excite a transition with this energy difference, the wavelength of incident photons must be

$$\lambda = \frac{hc}{\Delta E_{\text{vib}}} = \frac{1\ 240\ \text{eV}\cdot\text{nm}}{0.359\ \text{eV}} = 3.45\times10^3\ \text{nm}$$

The incident photons have a wavelength longer than this, which means they have less energy than 0.359 eV. Therefore, these photons cannot excite the molecule to the first excited state.

P43.13 The mass to consider is the molecule's reduced mass. Iodine has atomic mass 126.90 u and a hydrogen atom is 1.007 9 u, so the reduced mass of HI is

$$\mu = \frac{m_1 m_2}{m_1 + m_2} = \frac{(126.90\ \text{u})(1.007\ 9\ \text{u})}{126.90\ \text{u} + 1.007\ 9\ \text{u}} = 0.999\ 96\ \text{u}$$

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Now for the energy of the ground state we have

$$E = \frac{1}{2}kA^2 = \left(0 + \frac{1}{2}\right)hf = \frac{1}{2}\frac{h}{2\pi}\sqrt{\frac{k}{\mu}}$$

So the amplitude is

$$A = \sqrt{\frac{h}{2\pi k}\sqrt{\frac{k}{\mu}}} = \left(\frac{h}{2\pi}\right)^{1/2}\left(\frac{1}{k\mu}\right)^{1/4}$$

(a) For HI we have

$$A = \left(\frac{6.626\times10^{-34}\,\text{J}\cdot\text{s}}{2\pi}\right)^{1/2} \times\left(\frac{1}{(320\ \text{N/m})(0.999\ 96)(1.66\times10^{-27}\,\text{kg})}\right)^{1/4}$$

$$= \boxed{1.20\times10^{-11}\ \text{m} = 12.0\ \text{pm}}$$

(b) Flourine has an atomic mass of 18.998 4 u, so, for HF,

$$\mu = \frac{m_1 m_2}{m_1 + m_2} = \frac{(18.998\ 4\ \text{u})(1.007\ 9\ \text{u})}{18.998\ 4\ \text{u} + 1.007\ 9\ \text{u}} = 0.957\ 12\ \text{u}$$

and

$$A = \left(\frac{6.626\times10^{-34}\,\text{J}\cdot\text{s}}{2\pi}\right)^{1/2} \times\left(\frac{1}{(970\ \text{N/m})(0.957\ 12)(1.66\times10^{-27}\,\text{kg})}\right)^{1/4}$$

$$= \boxed{9.22\times10^{-12}\,\text{m} = 9.22\ \text{pm}}$$

P43.14 The energy of a rotational transition is $\Delta E = \left(\frac{\hbar^2}{I}\right)J$, where J is the rotational quantum number of the higher energy state (see Equation 43.7). We do not know J from the data. However,

$$\Delta E = \frac{hc}{\lambda} = \frac{\left(6.626\times10^{-34}\ \text{J}\cdot\text{s}\right)\left(2.998\times10^{8}\ \text{m/s}\right)}{\lambda}\left(\frac{1\ \text{eV}}{1.602\times10^{-19}\ \text{J}}\right)$$

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For each observed wavelength,

λ (mm)	E (eV)
0.120 4	0.010 30
0.096 4	0.012 86
0.080 4	0.015 42
0.069 0	0.017 97
0.060 4	0.020 53

The $\Delta E's$ consistently increase by 0.002 56 eV.

$$E_1 = \frac{\hbar^2}{I} = 0.002\ 56\ \text{eV}$$

and

$$I = \frac{\hbar^2}{E_1} = \frac{\left(1.055\times10^{-34}\ \text{J}\cdot\text{s}\right)^2}{\left(0.002\ 56\ \text{eV}\right)}\left(\frac{1\ \text{eV}}{1.60\times10^{-19}\ \text{J}}\right)$$
$$= \boxed{2.72\times10^{-47}\ \text{kg}\cdot\text{m}^2}$$

For the HCl molecule, the internuclear radius is

$$r = \sqrt{\frac{I}{\mu}} = \sqrt{\frac{2.72\times10^{-47}}{1.62\times10^{-27}}}\ \text{m} = 0.130\ \text{nm}$$

P43.15 (a) The reduced mass of NaCl is

$$\mu = \frac{m_{\text{Na}}m_{\text{Cl}}}{m_{\text{Na}}+m_{\text{Cl}}} = \frac{(22.99\ \text{u})(35.45\ \text{u})}{22.99\ \text{u}+35.45\ \text{u}}\left(\frac{1.66\times10^{-27}\ \text{kg}}{\text{u}}\right)$$
$$= \boxed{2.32\times10^{-26}\ \text{kg}}$$

(b) Its moment of inertia is

$$I = \mu r^2 = \left(2.32\times10^{-26}\ \text{kg}\right)\left(0.280\times10^{-9}\ \text{m}\right)^2$$
$$= \boxed{1.82\times10^{-45}\ \text{kg}\cdot\text{m}^2}$$

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(c) The wavelength of the emitted photon is found from:

$$\frac{hc}{\lambda} = \Delta E = \frac{\hbar^2}{2I}2(2+1) - \frac{\hbar^2}{2I}1(1+1) = \frac{3\hbar^2}{I} - \frac{\hbar^2}{I} = \frac{2\hbar^2}{I} = \frac{2h^2}{4\pi^2 I}$$

then,

$$\lambda = \frac{c4\pi^2 I}{2h} = \frac{\left(3.00\times10^8 \text{ m/s}\right)4\pi^2\left(1.82\times10^{-45} \text{ kg}\cdot\text{m}^2\right)}{2\left(6.626\times10^{-34} \text{ J}\cdot\text{s}\right)}$$

$$= \boxed{1.62 \text{ cm}}$$

P43.16 Masses m_1 and m_2 have the respective distances r_1 and r_2 from the center of mass. Then,

$$m_1 r_1 = m_2 r_2 \quad \text{and} \quad r_1 + r_2 = r$$

So, $$r_1 = \frac{m_2 r_2}{m_1}$$

and thus, $$\frac{m_2 r_2}{m_1} + r_2 = r \quad \rightarrow \quad r_2 = \frac{m_1 r}{m_1 + m_2}$$

Also, $$r_2 = \frac{m_1 r_1}{m_2}$$

thus, $$r_1 + \frac{m_1 r_1}{m_2} = r \quad \rightarrow \quad r_1 = \frac{m_2 r}{m_1 + m_2}$$

The moment of inertia of the molecule is then

$$I = m_1 r_1^2 + m_2 r_2^2 = m_1 \frac{m_2^2 r^2}{\left(m_1 + m_2\right)^2} + m_2 \frac{m_1^2 r^2}{\left(m_1 + m_2\right)^2}$$

$$= \frac{m_1 m_2\left(m_2 + m_1\right)r^2}{\left(m_1 + m_2\right)^2} = \frac{m_1 m_2 r^2}{m_1 + m_2} = \boxed{\mu r^2}$$

P43.17 (a) The reduced mass of the O_2 is

$$\mu = \frac{m_O m_O}{m_O + m_O} = \frac{(16.00 \text{ u})(16.00 \text{ u})}{(16.00 \text{ u}) + (16.00 \text{ u})} = 8 \text{ u}$$

$$= 8\left(1.66\times10^{-27} \text{ kg}\right) = 1.33\times10^{-26} \text{ kg}$$

The moment of inertia is then

$$I = \mu r^2 = \left(1.33\times10^{-26} \text{ kg}\right)\left(1.20\times10^{-10} \text{ m}\right)^2$$

$$= 1.91\times10^{-46} \text{ kg}\cdot\text{m}^2$$

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The rotational energies are

$$E_{\text{rot}} = \frac{\hbar^2}{2I} J(J+1) = \frac{\left(6.626\times10^{-34}\ \text{J}\cdot\text{s}/2\pi\right)^2}{2\left(1.91\times10^{-46}\ \text{kg}\cdot\text{m}^2\right)} J(J+1)$$

Thus,

$$E_{\text{rot}} = \left(2.91\times10^{-23}\ \text{J}\right) J(J+1)$$

and for $J = 0, 1, 2,$

$$E_{\text{rot}} = \boxed{0,\ 3.63\times10^{-4}\ \text{eV},\ 1.09\times10^{-3}\ \text{eV}}$$

(b) The vibrational energies are given by

$$\begin{aligned} E_{\text{vib}} &= \left(v+\frac{1}{2}\right)\hbar\sqrt{\frac{k}{\mu}} \\ &= \left(v+\frac{1}{2}\right)\left(\frac{6.626\times10^{-34}\ \text{J}\cdot\text{s}}{2\pi}\right)\sqrt{\frac{1\,177\ \text{N/m}}{8\left(1.66\times10^{-27}\ \text{kg}\right)}} \\ &= \left(v+\frac{1}{2}\right)\left(3.14\times10^{-20}\ \text{J}\right)\left(\frac{1\ \text{eV}}{1.602\times10^{-19}\ \text{J}}\right) \\ &= \left(v+\frac{1}{2}\right)(0.196\ \text{eV}) \end{aligned}$$

For $v = 0, 1, 2,$ $\boxed{E_{\text{vib}} = 0.098\,0\ \text{eV}, 0.294\ \text{eV}, 0.490\ \text{eV}}$.

P43.18 (a) In benzene, the dashed lines form equilateral triangles, so the carbon atoms are each 0.110 nm from the axis and each hydrogen atom is (0.110 + 0.100 nm) = 0.210 nm from the axis. Thus, the moment of inertia is given by

$$\begin{aligned} I = \sum mr^2 &= 6\left(1.99\times10^{-26}\ \text{kg}\right)\left(0.110\times10^{-9}\ \text{m}\right)^2 \\ &\quad + 6\left(1.67\times10^{-27}\ \text{kg}\right)\left(0.210\times10^{-9}\ \text{m}\right)^2 \\ &= \boxed{1.89\times10^{-45}\ \text{kg}\cdot\text{m}^2} \end{aligned}$$

(b) The allowed rotational energies are then

$$\begin{aligned} E_{\text{rot}} &= \frac{\hbar^2}{2I} J(J+1) = \frac{\left(1.055\times10^{-34}\ \text{J}\cdot\text{s}\right)^2}{2\left(1.89\times10^{-45}\ \text{kg}\cdot\text{m}^2\right)} J(J+1) \\ &= \left(2.95\times10^{-24}\ \text{J}\right) J(J+1) = \left(18.4\times10^{-6}\ \text{eV}\right) J(J+1) \end{aligned}$$

$\boxed{E_{\text{rot}} = 18.4J(J+1), \text{ where } E_{\text{rot}} \text{ is in microelectron volts and } J = 0,1,2,3,\ldots.}$

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The first five of these allowed energies are:

$$E_{\text{rot}} = 0,\ 36.9\ \mu\text{eV},\ 111\ \mu\text{eV},\ 221\ \mu\text{eV},\ \text{and}\ 369\ \mu\text{eV}$$

P43.19 We carry extra digits through the solution because part (c) involves the subtraction of two close numbers. The longest wavelength corresponds to the smallest energy difference between the rotational energy levels. It is between $J = 0$ and $J = 1$, namely

$$\Delta E_{\text{min}} = \frac{\hbar^2}{I}$$

The wavelength is then

$$\lambda = \frac{hc}{\Delta E_{\text{min}}} = \frac{hc}{\hbar^2/I} = \frac{4\pi^2 Ic}{h}$$

If $\mu = \dfrac{m_{\text{H}} m_{\text{Cl}}}{m_{\text{H}} + m_{\text{Cl}}}$ is the reduced mass, then

$$I = \mu r^2 = \mu\left(0.127\ 46 \times 10^{-9}\ \text{m}\right)^2$$

and therefore,

$$\lambda = \frac{4\pi^2\left[\mu\left(0.127\ 46 \times 10^{-9}\ \text{m}\right)^2\right]\left(2.997\ 925 \times 10^{8}\ \text{m/s}\right)}{6.626\ 075 \times 10^{-34}\ \text{J}\cdot\text{s}}$$

$$= \left(2.901\ 830 \times 10^{23}\ \text{m/kg}\right)\mu \qquad \textbf{[1]}$$

(a) For ^{35}Cl,

$$\mu_{35} = \frac{\left(1.007\ 825\text{u}\right)\left(34.968\ 853\text{u}\right)}{1.007\ 825\text{u} + 34.968\ 853\text{u}}\left(\frac{1.660\,540 \times 10^{-27}\ \text{kg}}{\text{u}}\right)$$

$$= 1.626\ 653 \times 10^{-27}\ \text{kg}$$

From equation [1],

$$\lambda_{35} = \left(2.901\ 830 \times 10^{23}\ \text{m/kg}\right)\left(1.626\ 653 \times 10^{-27}\ \text{kg}\right)$$

$$= \boxed{472\ \mu\text{m}}$$

(b) For ^{37}Cl,

$$\mu_{37} = \frac{\left(1.007\ 825\text{u}\right)\left(36.965\ 903\text{u}\right)}{1.007\ 825\text{u} + 36.965\ 903\text{u}}\left(\frac{1.660\,540 \times 10^{-27}\ \text{kg}}{\text{u}}\right)$$

$$= 1.629\ 118 \times 10^{-27}\ \text{kg}$$

From equation [1],

$$\lambda_{37} = \left(2.901\ 830 \times 10^{23}\ \text{m/kg}\right)\left(1.629\ 118 \times 10^{-27}\ \text{kg}\right) = \boxed{473\ \mu\text{m}}$$

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(c) The separation in wavelength is

$$\lambda_{37} - \lambda_{35} = 472.742\ 4\ \mu\text{m} - 472.027\ 0\ \mu\text{m} = \boxed{0.715\ \mu\text{m}}$$

P43.20 We find an average spacing between peaks by counting 22 gaps (counting the central gap as two) between 7.96×10^{13} Hz and 9.24×10^{13} Hz:

$$\Delta f = \frac{(9.24 - 7.96) \times 10^{13}\ \text{Hz}}{22} = 0.058\,2 \times 10^{13}\ \text{Hz}$$

$$= 5.82 \times 10^{11}\,\text{Hz} = \frac{1}{h}\left(\frac{h^2}{4\pi^2 I}\right)$$

The moment of inertia is then

$$I = \frac{h}{4\pi^2 \Delta f} = \frac{6.626 \times 10^{-34}\ \text{J}\cdot\text{s}}{4\pi^2 \left(5.82 \times 10^{11}\ \text{s}^{-1}\right)} = \boxed{2.88 \times 10^{-47}\ \text{kg}\cdot\text{m}^2}$$

P43.21 We carry extra digits through the solution because the given wavelengths are close together.

(a) The energy levels are given by

$$E_{vJ} = \left(v + \frac{1}{2}\right)hf + \frac{\hbar^2}{2I}J(J+1)$$

Therefore,

$$E_{00} = \frac{1}{2}hf,\ E_{11} = \frac{3}{2}hf + \frac{\hbar^2}{I},\ \text{and}\ E_{02} = \frac{1}{2}hf + \frac{3\hbar^2}{I}$$

Then,

$$\Delta E_1 = E_{11} - E_{00} = hf + \frac{\hbar^2}{I} = \frac{hc}{\lambda_1}$$

$$= \frac{\left(6.626\,075 \times 10^{-34}\ \text{J}\cdot\text{s}\right)\left(2.997\,925 \times 10^{8}\ \text{m/s}\right)}{2.211\,2 \times 10^{-6}\ \text{m}}$$

$$\Delta E_1 = hf + \frac{\hbar^2}{I} = 8.983\,573 \times 10^{-20}\ \text{J} \qquad \textbf{[1]}$$

and

$$\Delta E_2 = E_{11} - E_{02} = hf - \frac{2\hbar^2}{I} = \frac{hc}{\lambda_2}$$

$$= \frac{\left(6.626\,075 \times 10^{-34}\ \text{J}\cdot\text{s}\right)\left(2.997\,925 \times 10^{8}\ \text{m/s}\right)}{2.405\,4 \times 10^{-6}\ \text{m}}$$

$$\Delta E_2 = hf - \frac{2\hbar^2}{I} = 8.258\,284 \times 10^{-20}\ \text{J} \qquad \textbf{[2]}$$

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Subtracting equation [2] from [1] gives,

$$\Delta E_1 - \Delta E_2 = \left(hf + \frac{\hbar^2}{I} \right) - \left(hf - \frac{2\hbar^2}{I} \right) = \frac{hc}{\lambda_1} - \frac{hc}{\lambda_2}$$

solving,

$$\frac{3\hbar^2}{I} = hc\left(\frac{1}{\lambda_1} - \frac{1}{\lambda_2} \right) \quad \rightarrow \quad \frac{3h}{4\pi^2 I} = c\left(\frac{1}{\lambda_1} - \frac{1}{\lambda_2} \right)$$

Then,

$$\begin{aligned} \frac{3h}{4\pi^2 I} &= c\left(\frac{1}{\lambda_1} - \frac{1}{\lambda_2} \right) \\ &= \left(2.997\,925 \times 10^8 \text{ m/s} \right) \\ &\qquad \times \left(\frac{1}{2.211\,2 \times 10^{-6} \text{ m}} - \frac{1}{2.405\,4 \times 10^{-6} \text{ m}} \right) \\ &= 1.0946 \times 10^{13} \text{ s}^{-1} \end{aligned}$$

Solving for the moment of inertia then gives

$$I = \frac{3\left(6.626\,075 \times 10^{-34} \text{ J} \cdot \text{s} \right)}{4\pi^2 \left(1.094\,6 \times 10^{13} \text{ s}^{-1} \right)} = \boxed{4.60 \times 10^{-48} \text{ kg} \cdot \text{m}^2}$$

(b) From equation [1]:

$$\begin{aligned} f_1 &= \frac{\Delta E_1}{h} - \frac{\hbar^2}{2\pi I} \\ &= \frac{8.983\,573 \times 10^{-20} \text{ J}}{6.626\,075 \times 10^{-34} \text{ J} \cdot \text{s}} - \frac{\left(6.626\,075 \times 10^{-34} \text{ J} \cdot \text{s} \right)}{2\pi \left(4.600\,060 \times 10^{-48} \text{ kg} \cdot \text{m}^2 \right)} \\ &= \boxed{1.32 \times 10^{14} \text{ Hz}} \end{aligned}$$

(c) The moment of inertia of the molecule is given by $I = \mu r^2$, where μ is the reduced mass,

$$\mu = \frac{1}{2} m_H = \frac{1}{2}\left(1.007\,825\text{u} \right) = 8.367\,669 \times 10^{-28} \text{ kg}$$

The equilibrium separation distance is then,

$$r = \sqrt{\frac{I}{\mu}} = \sqrt{\frac{4.600\,060 \times 10^{-48} \text{ kg} \cdot \text{m}^2}{8.367\,669 \times 10^{-28} \text{ kg}}} = \boxed{0.074\,1 \text{ nm}}$$

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P43.22 The emission energies are the same as the absorption energies, but the final state must be below ($v = 1, J = 0$). The transition must satisfy $\Delta J = \pm 1,$ so it must end with $J = 1$. To be lower in energy, the state must be ($v = 0, J = 1$). The emitted photon energy is therefore

$$hf_{\text{photon}} = \left(E_{\text{vib}}\big|_{v=1} + E_{\text{rot}}\big|_{J=0}\right) - \left(E_{\text{vib}}\big|_{v=0} + E_{\text{rot}}\big|_{J=1}\right)$$

$$= \left(E_{\text{vib}}\big|_{v=1} - E_{\text{vib}}\big|_{v=0}\right) - \left(E_{\text{rot}}\big|_{J=1} - E_{\text{rot}}\big|_{J=0}\right)$$

$$hf_{\text{photon}} = hf_{\text{vib}} - hf_{\text{rot}}$$

Thus, $f_{\text{photon}} = f_{\text{vib}} - f_{\text{rot}} = 6.42 \times 10^{13}\ \text{Hz} - 1.15 \times 10^{11}\ \text{Hz}$

$$= 6.41 \times 10^{13}\ \text{Hz} = \boxed{64.1\ \text{THz}}$$

P43.23 The moment of inertia about the molecular axis is

$$I_y = \frac{2}{5}mr^2 + \frac{2}{5}mr^2 = \frac{4}{5}m\left(2.00 \times 10^{-15}\ \text{m}\right)^2$$

The moment of inertia about a perpendicular axis is

$$I_x = m\left(\frac{R}{2}\right)^2 + m\left(\frac{R}{2}\right)^2 = \frac{m}{2}\left(2.00 \times 10^{-10}\ \text{m}\right)^2$$

The allowed rotational energies are $E_{\text{rot}} = \left(\frac{\hbar^2}{2I}\right)J(J+1)$, so the energy of the first excited state is $E_1 = \frac{\hbar^2}{I}$. The ratio is therefore

$$\frac{E_{1,\,y}}{E_{1,\,x}} = \frac{\left(\hbar^2/I_y\right)}{\left(\hbar^2/I_x\right)} = \frac{I_x}{I_y} = \frac{(1/2)m\left(2.00 \times 10^{-10}\ \text{m}\right)^2}{(4/5)m\left(2.00 \times 10^{-15}\ \text{m}\right)^2}$$

$$= \frac{5}{8}\left(10^5\right)^2 = \boxed{6.25 \times 10^9}$$

Section 43.3 Bonding in Solids

P43.24 (a) Consider a cubical salt crystal of edge length 0.1 mm.

The number of atoms is $\left(\frac{10^{-4}\ \text{m}}{0.261 \times 10^{-9}\ \text{m}}\right)^3 = 6 \times 10^{16}$ $\boxed{\sim 10^{17}}$.

(b) This number of salt crystals would have volume

$$\left(10^{-4}\ \text{m}\right)^3 6 \times 10^{16} = 6 \times 10^4\ \boxed{\sim 10^5\ \text{m}^3}$$

If it is cubic, it has edge length 40 m.

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P43.25 The ionic cohesive energy is

$$U = -\frac{\alpha k_e e^2}{r_0}\left(1-\frac{1}{m}\right)$$

$$= -(1.747\,6)\left(8.99\times10^{9}\right)\frac{\left(1.60\times10^{-19}\right)^2}{\left(0.281\times10^{-9}\right)}\left(1-\frac{1}{8}\right)$$

$$= -1.25\times10^{-18}\ \text{J} = \boxed{-7.83\ \text{eV}}$$

P43.26 We assume the ions are all singly ionized. The total potential energy is obtained by summing over all interactions of our ion with others:

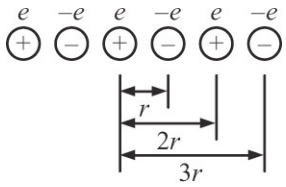
$$U = \sum_{i\neq j} k_e \frac{q_i q_j}{r_{ij}}$$

$$= -k_e\left[\frac{e^2}{r}+\frac{e^2}{r}-\frac{e^2}{2r}-\frac{e^2}{2r}+\frac{e^2}{3r}+\frac{e^2}{3r}-\frac{e^2}{4r}-\frac{e^2}{4r}+\cdots\right]$$

$$U = -2k_e\frac{e^2}{r}\left[1-\tfrac{1}{2}+\tfrac{1}{3}-\tfrac{1}{4}+\cdots\right]$$

But from Appendix B.5,

$$\ln(1+x) = x-\frac{x^2}{2}+\frac{x^3}{3}-\frac{x^4}{4}+\cdots$$



ANS. FIG. P43.26

Our series follows this pattern with $x = 1$, so the potential energy of one ion due to its interactions with all the others is

$$U = (-2\ln 2)k_e\frac{e^2}{r} = \boxed{-k_e\alpha\frac{e^2}{r}\ \text{where}\ \alpha = 2\ln 2}$$

Section 43.4 Free-Electron Theory of Metals

Section 43.5 Band Theory of Solids

P43.27 Taking $E_F = 5.48$ eV for sodium at 800 K,

$$f(E) = \frac{1}{e^{(E-E_F)/k_BT}+1} \qquad \text{so} \qquad e^{(E-E_F)/k_BT} = \frac{1}{f(E)}-1$$

Then, $$\frac{E-E_F}{k_BT} = \ln\left(\frac{1}{f(E)}-1\right)$$

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and $$E = E_F + k_B T \ln\left(\frac{1}{f(E)} - 1\right)$$

substituting numerical values,

$$E = 5.48 \text{ eV} + \left(1.38 \times 10^{-23} \text{ J/K}\right)$$
$$\times \left(1.602 \times 10^{-19} \text{ eV/J}\right)(800 \text{ K}) \ln\left(\frac{1}{0.950} - 1\right) = \boxed{5.28 \text{ eV}}$$

P43.28 (a) The Fermi energy is proportional to the spatial concentration of free electrons to the two-thirds power.

(b) From Equation 43.25,

$$E_F = \frac{h^2}{2m}\left(\frac{3n_e}{8\pi}\right)^{2/3}$$
$$= \frac{\left(6.626 \times 10^{-34} \text{ J} \cdot \text{s}\right)^2}{2\left(9.11 \times 10^{-31} \text{ kg}\right)\left(1.60 \times 10^{-19} \text{ J/eV}\right)}\left(\frac{3}{8\pi}\right)^{2/3} n_e^{2/3}$$

becomes

$$E_F = \left(3.65 \times 10^{-19}\right) n_e^{2/3}$$

where E_F is in electron volts and n_e in electrons per cubic meter.

(c) Copper has the greater concentration of free electrons by a factor of

$$\frac{n_e(\text{Cu})}{n_e(\text{K})} = \frac{8.46 \times 10^{-19} \text{ m}^{-3}}{1.40 \times 10^{-19} \text{ m}^{-3}} = \boxed{6.04}$$

(d) Copper has the greater Fermi energy, 7.05 eV.

(e) The Fermi energy is larger by a factor of 7.05 eV/2.12 eV = 0.333.

(f) This behavior agrees with the proportionality because $E_F \sim n_e^{2/3}$ and $6.04^{2/3} = 3.32$.

P43.29 The melting point of silver is 1 234 K. Its Fermi energy at 300 K is 5.48 eV. The approximate fraction of electrons excited is

$$\frac{k_B T}{E_F} = \frac{\left(1.38 \times 10^{-23} \text{ J/K}\right)\left(1\,234 \text{ K}\right)}{\left(5.48 \text{ eV}\right)\left(1.60 \times 10^{-19} \text{ J/eV}\right)} \approx \boxed{2\%}$$

P43.30 (a) Setting the kinetic energy equal to the Fermi energy,

$$\frac{1}{2} m v^2 = 7.05 \text{ eV}$$

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we solve for the speed of the conduction electron as

$$v = \sqrt{\frac{2(7.05\text{ eV})\left(1.60\times10^{-19}\text{ J/eV}\right)}{9.11\times10^{-31}\text{ kg}}}$$

$$= 1.57\times10^{6}\text{ m/s} = \boxed{1.57\text{ Mm/s}}$$

(b) Compared to the drift velocity of 0.1 mm/s = 10^{-4} mm/s, the speed is larger by ten orders of magnitude. The energy of an electron at room temperature is typically $k_B T = \frac{1}{40}$ eV.

P43.31 (a) From Equation 43.26,

$$E_{\text{avg}} = \frac{3}{5}E_F = 0.6(7.05\text{ eV}) = \boxed{4.23\text{ eV}}$$

(b) The average energy of a molecule in an ideal gas is $\frac{3}{2}k_B T$ so we have

$$T = \frac{2}{3}\frac{4.23\text{ eV}}{1.38\times10^{-23}\text{ J/K}}\frac{1.6\times10^{-19}\text{ J}}{1\text{ eV}} = \boxed{3.27\times10^{4}\text{ K}}$$

P43.32 For edge d = 1.00 mm,

$$V = d^3 = \left(1.00\times10^{-3}\text{ m}\right)^3 = 1.00\times10^{-9}\text{ m}^3$$

The density of states is

$$g(E) = CE^{1/2} = \frac{8\sqrt{2}\pi m_e^{3/2}}{h^3}E^{1/2}$$

or

$$g(E) = \frac{8\sqrt{2}\pi\left(9.11\times10^{-31}\text{ kg}\right)^{3/2}}{\left(6.626\times10^{-34}\text{ J}\cdot\text{s}\right)^3}\sqrt{(4.00\text{ eV})\left(1.60\times10^{-19}\text{ J/eV}\right)}$$

$$g(E) = 8.50\times10^{46}\text{ m}^{-3}\cdot\text{J}^{-1} = 1.36\times10^{28}\text{ m}^{-3}\cdot\text{eV}^{-1}$$

So, the total number of electrons is

$$N = \left[g(E)\right](\Delta E)V$$

$$= \left(1.36\times10^{28}\text{ m}^{-3}\cdot\text{eV}^{-1}\right)\left(0.025\,0\text{ eV}\right)\left(1.00\times10^{-9}\text{ m}^3\right)$$

$$= \boxed{3.40\times10^{17}\text{ electrons}}$$

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P43.33 For sodium, $M = 23.0\ \text{g/mol}$ and $\rho = 0.971\ \text{g/cm}^3$. Sodium contributes one electron per atom to the conduction band.

(a) The density of conduction electrons is

$$n_e = \frac{N_A \rho}{M} = \frac{\left(6.02 \times 10^{23}\ \text{electrons/mol}\right)\left(0.971\ \text{g/cm}^3\right)}{23.0\ \text{g/mol}}$$

$$n_e = 2.54 \times 10^{22}\ \text{electrons/cm}^3 = \boxed{2.54 \times 10^{28}\ \text{m}^{-3}}$$

(b) From Equation 43.25,

$$E_F = \left(\frac{h^2}{2m}\right)\left(\frac{3n_e}{8\pi}\right)^{2/3}$$

$$= \frac{\left(6.626 \times 10^{-34}\ \text{J}\cdot\text{s}\right)^2}{2\left(9.11 \times 10^{-31}\ \text{kg}\right)}\left[\frac{3\left(2.54 \times 10^{28}\ \text{m}^{-3}\right)}{8\pi}\right]^{2/3}$$

$$= 5.05 \times 10^{-19}\ \text{J} = \boxed{3.15\ \text{eV}}$$

P43.34 From Equation 43.24, the number density of free electrons is

$$n_e = \frac{2}{3}\frac{8\sqrt{2}\pi m_e^{3/2}}{h^3}E_F^{3/2}$$

$$= \frac{2}{3}\frac{8\sqrt{2}\pi\left(9.11 \times 10^{-31}\text{kg}\right)^{3/2}}{\left(6.626 \times 10^{-34}\text{J}\cdot\text{s}\right)^3}\left(5.48\text{eV}\right)^{3/2}\left(\frac{1.602 \times 10^{-19}\text{J}}{1\ \text{eV}}\right)^{3/2}$$

$$= 5.83 \times 10^{28}\text{m}^{-3}$$

Then, the number density of atoms in the metal is

$$n_{\text{atoms}} = \frac{nN_A}{V} = \frac{mN_A}{MV} = \frac{\rho N_A}{M}$$

$$= \frac{\left(4.90 \times 10^3\ \text{kg/m}^3\right)\left(6.02 \times 10^{23}\ \text{mol}^{-1}\right)}{0.100\ \text{kg/mol}}$$

$$= 2.95 \times 10^{28}\ \text{m}^{-3}$$

Then the number of free electrons per atom is

$$\frac{n_e}{n_{\text{atoms}}} = \frac{5.83 \times 10^{28}\ \text{m}^{-3}}{2.95 \times 10^{28}\ \text{m}^{-3}} = 1.97$$

Therefore, there are approximately two free electrons per atom for this metal, not one.

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P43.35 From Table 43.2, the Fermi energy for copper at 300 K is 7.05 eV. From Equation 43.19, the Fermi-Dirac distribution function, the occupation probability is

$$f(E) = \frac{1}{e^{(E-E_F)/k_BT} + 1} = \frac{1}{e^{(0.99E_F - E_F)/k_BT} + 1}$$

$$= \frac{1}{e^{\left[-0.01(7.05\text{ eV})(1.602\times10^{-19}\text{ J/eV})\right]/\left[(1.381\times10^{-23}\text{ J/K})(300\text{ K})\right]} + 1}$$

$$= \frac{1}{e^{-2.72} + 1} = \boxed{0.939}$$

P43.36 From Equation 43.19, the Fermi-Dirac distribution function, the occupation probability is

$$f(E) = \frac{1}{e^{(E-E_F)/k_BT} + 1} = \frac{1}{e^{(\beta E_F - E_F)/k_BT} + 1} = \boxed{\frac{1}{e^{(\beta-1)E_F/k_BT} + 1}}$$

P43.37 Consider first the wave function in x. At $x = 0$ and $x = L$, $\psi = 0$. Therefore,

$$\sin k_x L = 0 \quad \text{and} \quad k_x L = \pi, 2\pi, 3\pi, \ldots$$

Similarly, $\sin k_y L = 0 \quad \text{and} \quad k_y L = \pi, 2\pi, 3\pi, \ldots$

and $\sin k_z L = 0 \quad \text{and} \quad k_z L = \pi, 2\pi, 3\pi, \ldots$

Then,

$$\psi = A \sin\left(\frac{n_x \pi x}{L}\right) \sin\left(\frac{n_y \pi y}{L}\right) \sin\left(\frac{n_z \pi z}{L}\right)$$

From Schrödinger's Equation, $\frac{\partial^2\psi}{\partial x^2} + \frac{\partial^2\psi}{\partial y^2} + \frac{\partial^2\psi}{\partial z^2} = \frac{2m_e}{\hbar^2}(U - E)\psi$, we have inside the box, where $U = 0$,

$$\left(-\frac{n_x^2\pi^2}{L^2} - \frac{n_y^2\pi^2}{L^2} - \frac{n_z^2\pi^2}{L^2}\right)\psi = \frac{2m_e}{\hbar^2}(-E)\psi$$

Therefore,

$$\boxed{E = \frac{\hbar^2\pi^2}{2m_e L^2}\left(n_x^2 + n_y^2 + n_z^2\right) \qquad n_x,\ n_y,\ n_z = 1,\ 2,\ 3,\ \ldots}$$

Outside the box we require $\psi = 0$. The minimum energy state inside the box is $n_x = n_y = n_z = 1$, with $E = \frac{3\hbar^2\pi^2}{2m_e L^2}$.

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P43.38 The density of states at the energy E is $g(E) = CE^{1/2}$.

(a) Hence, the required ratio is

$$R_{states} = \frac{g(8.50\text{ eV})}{g(7.05\text{ eV})} = \frac{C(8.50)^{1/2}}{C(7.05)^{1/2}} = \boxed{1.10}$$

(b) From Equation 43.22, we see that the number of occupied states between energy E and energy $E + dE$ is

$$N(E)dE = \frac{CE^{1/2}}{e^{(E-E_F)/k_BT} + 1}dE$$

Hence, the required ratio is

$$R_{occupied\ states} = \frac{N(8.50\text{ eV})}{N(7.05\text{ eV})} = \sqrt{\frac{8.50}{7.05}}\left[\frac{e^{(7.05-7.05)/k_BT}+1}{e^{(8.50-7.05)/k_BT}+1}\right]$$

At $T = 300$ K, we compute

$$k_BT = \left(1.380\ 65\times10^{-23}\frac{\text{J}}{\text{K}}\right)(300.000\text{ K})\left(\frac{1\text{ eV}}{1.602\ 18\times10^{-19}\text{J}}\right)$$
$$= 0.025\ 852\ 0\text{ eV}$$

so $$R_{occ.st} = \left(\frac{8.50}{7.05}\right)^{1/2}\left(\frac{2}{e^{1.45/0.025\ 852\ 0}+1}\right) = \boxed{9.61\times10^{-25}}$$

With an exponent of 56.1, the derivative of the exponential function is so large that none of the digits in 9.61 is really significant. Different-looking answers would result from different choices of how precisely to represent the input data.

(c) The answer to part (b) is vastly smaller than the answer to (a). Very few states well above the Fermi energy are occupied at room temperature.

P43.39 We are to compute

$$E_{avg} = \frac{1}{n_e}\int_0^\infty EN(E)\,dE$$

where from Equation 43.22,

$$N(E) = \frac{CE^{1/2}}{e^{(E-E_F)/k_BT}+1} = Cf(E)E^{1/2}$$

with $$C = \frac{8\sqrt{2}\pi m_e^{3/2}}{h^3}$$

But at $T = 0$ the Fermi-Dirac distribution function is $f(E) = 0$ for $E > E_F$, and $f(E) = 1$ for $E < E_F$. So we can take $N(E) = CE^{1/2}$ just for energies

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up to the Fermi energy. The average we want is then

$$E_{\text{avg}} = \frac{1}{n_e}\int_0^{E_F} CE^{3/2}dE = \frac{2C}{5n_e}E_F^{5/2}$$

But from Equation 43.24, $\frac{C}{n_e} = \frac{3}{2}E_F^{-3/2}$, so

$$E_{\text{avg}} = \left(\frac{2}{5}\right)\left(\frac{3}{2}\right)\left(E_F^{-3/2}\right)E_F^{5/2} = \boxed{\frac{3}{5}E_F}$$

Section 43.6 Electrical Conduction in Metals, Insulators, and Semiconductors

P43.40 (a) If $\lambda \le 1.00\ \mu\text{m} = 1.00\times10^3$ nm, then photons of sunlight have energy

$$E \ge \frac{hc}{\lambda_{\max}} = \frac{1240\ \text{eV}\cdot\text{nm}}{1.00\times10^3\ \text{m}} = 1.24\ \text{eV}$$

The gap should be less than or equal to 1.24 eV.

(b) Because silicon has an energy gap of 1.14 eV, it can absorb the energy of nearly all of the photons in sunlight and is an appropriate material for a solar energy collector.

P43.41 (a) E_g = 1.14 eV for Si. The photon energy, given by $E = hf$, must be at least this energy. Then,

$$f = \frac{E}{h} = \frac{(1.14\ \text{eV})\left(1.602\times10^{-19}\ \text{J/eV}\right)}{6.626\times10^{-34}\ \text{J}\cdot\text{s}}$$

$$= 2.76\times10^{14}\ \text{Hz} = 276\times10^{12}\ \text{Hz} = \boxed{276\ \text{THz}}$$

(b) From $c = \lambda f$,

$$\lambda = \frac{c}{f} = \frac{3.00\times10^8\ \text{m/s}}{2.75\times10^{14}\ \text{Hz}} = 1.09\times10^{-6}\ \text{m} = \boxed{1.09\ \mu\text{m}}\ \text{(infrared)}$$

P43.42 (a) From Table 43.3, the energy gap for CdS is 2.42 eV, so photons of energy greater than 2.42 eV will be absorbed, corresponding to wavelengths shorter than

$$\lambda = \frac{hc}{E} = \frac{1240\ \text{eV}\cdot\text{nm}}{2.42\ \text{eV}} = 512\ \text{nm}$$

All the Balmer lines lie between the shortest (series limit)

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produced by the transition $n = \infty \to 2,$ with energy

$$\Delta E = 13.6 \text{ eV}\left(\frac{1}{2^2}\right) = 3.40 \text{ eV}$$

is $$\lambda = \frac{hc}{E} = \frac{1240 \text{ eV} \cdot \text{nm}}{3.40 \text{ eV}} = 365 \text{ nm}$$

The longest produced by the transition $n = 3 \to 2$, with energy

$$\Delta E = 13.6 \text{ eV}\left(\frac{1}{2^2} - \frac{1}{3^2}\right) = 1.89 \text{ eV}$$

is $$\lambda = \frac{hc}{E} = \frac{1240 \text{ eV} \cdot \text{nm}}{1.89 \text{ eV}} = 656 \text{ nm}$$

All the hydrogen Balmer lines except for the red line at 656 nm will be absorbed.

(b) The red line at 656 nm will be transmitted.

P43.43 The energy-bang gap is

$$E_g = \frac{hc}{\lambda} = \frac{1240 \text{ eV} \cdot \text{nm}}{650 \text{ nm}} \approx \boxed{1.91 \text{ eV}}$$

P43.44 The wavelength 0.512 μm = 512 nm. The corresponding photon energy is just sufficient to promote an electron across the gap.

$$E_g = \frac{hc}{\lambda} = \frac{1240 \text{ eV} \cdot \text{nm}}{512 \text{ nm}} = \boxed{2.42 \text{ eV}}$$

P43.45 If the photon energy is 5.47 eV or higher, the diamond window will absorb the photons. Here,

$$(hf)_{\max} = \frac{hc}{\lambda_{\min}} = 5.47 \text{ eV}$$

which gives

$$\lambda_{\min} = \frac{hc}{5.47 \text{ eV}} = \frac{1240 \text{ eV} \cdot \text{nm}}{5.47 \text{ eV}} = \boxed{227 \text{ nm}}$$

P43.46 (a) In the Bohr model we replace k_e by k_e/κ and m_e by m^*. Then the radius of the first Bohr orbit, $a_0 = \dfrac{\hbar^2}{m_e k_e e^2}$ in hydrogen, changes to

$$a' = \frac{\hbar^2}{m^*\left(k_e/\kappa\right)e^2} = \frac{\hbar^2 \kappa}{m^* k_e e^2} = \left(\frac{m_e}{m^*}\right)\kappa \frac{\hbar^2}{m_e k_e e^2} = \boxed{\left(\frac{m_e}{m^*}\right)\kappa a_0}$$

(b) Substituting numerical values,

$$a' = \left(\frac{m_e}{m^*}\right)\kappa a_0 = \left(\frac{m_e}{0.220 m_e}\right)(11.7)(0.052\,9 \text{ nm}) = \boxed{2.81 \text{ nm}}$$

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(c) The energy levels for hydrogen are $E_n = -\frac{k_e e^2}{2a_0}\frac{1}{n^2}$. Making the replacements $k_e \to k_e/\kappa$ and $a_0 \to a'$, we have

$$E_n' = -\frac{(k_e/\kappa)e^2}{2\left[(m_e/m^*)\kappa a_0\right]}\frac{1}{n^2} = -\frac{k_e e^2}{2\kappa^2\left[(m_e/m^*)a_0\right]}\frac{1}{n^2}$$

$$= -\frac{1}{\kappa^2}\left(\frac{m^*}{m_e}\right)\left(\frac{k_e e^2}{2a_0}\frac{1}{n^2}\right) = \boxed{-\left(\frac{m^*}{m_e}\right)\frac{E_n}{\kappa^2}}$$

(d) For $n = 1$,

$$E_1' = -0.220\left(\frac{13.6\text{ eV}}{11.7^2}\right) = \boxed{-0.021\,9\text{ eV}}.$$

Section 43.7 Semiconductor Devices

P43.47 Equation 43.27 is

$$I = I_0\left(e^{e\Delta V/k_B T} - 1\right)$$

Thus, $$e^{e(\Delta V)/k_B T} = 1 + \frac{I}{I_0}$$

and $$\Delta V = \frac{k_B T}{e}\ln\left(1 + \frac{I}{I_0}\right)$$

At $T = 300$ K,

$$\Delta V = \frac{(1.38\times10^{-23}\text{ J/K})(300\text{ K})}{1.60\times10^{-19}\text{ C}}\ln\left(1 + \frac{I}{I_0}\right)$$

$$= (25.9\text{ mV})\ln\left(1 + \frac{I}{I_0}\right)$$

(a) If $I = 9.00I_0$,

$$\Delta V = (25.9\text{ mV})\ln(10.0) = \boxed{59.5\text{ mV}}$$

(b) If $I = -0.900I_0$,

$$\Delta V = (25.9\text{ mV})\ln(0.100) = \boxed{-59.5\text{ mV}}$$

The basic idea behind a semiconductor device is that a large current or charge can be controlled by a small control voltage.

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P43.48 (a) The current in the diode, and thus in all elements of the series circuit, is $I = I_0\left(e^{e\Delta V/k_BT} - 1\right)$. Appling Kirchhoff's loop rule in the direction of the current, going through the negative to the positive side of the battery, then through the diode, and then the resistor, we get

$$\mathcal{E} - \Delta V - IR = 0$$

$$\mathcal{E} - \Delta V - I_0R\left(e^{e\Delta V/k_BT} - 1\right) = 0$$

or $$\mathcal{E} - \Delta V = I_0R\left(e^{e\Delta V/k_BT} - 1\right)$$

(b) The graphs to be plotted are the voltage across the resistor,

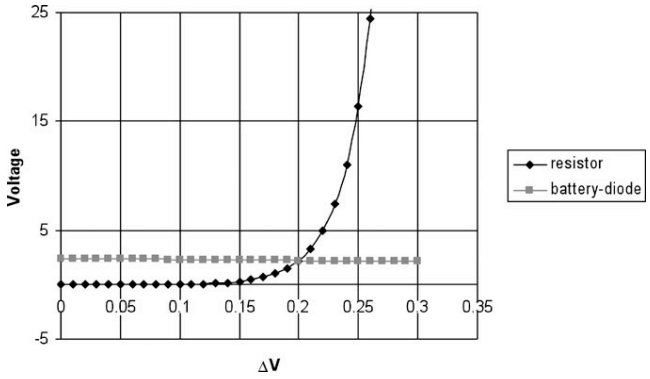
$$\Delta V_{\text{resistor}} = IR = \left(1.00\times10^{-6}\ \text{A}\right)(745\ \Omega)\left(e^{\Delta V/0.0250\ \text{V}} - 1\right)$$

$$= \left(7.45\times10^{-4}\ \text{A}\right)\left(e^{\Delta V/0.025\ \text{V}} - 1\right)$$

and the voltage across the battery and diode combined,

$$\Delta V_{\text{BD}} = \mathcal{E} - \Delta V = 2.42\ \text{V} - \Delta V$$

The graphs are plotted in ANS. FIG. P43.48 below.



ANS. FIG. P43.48

(c) The two graphs intersect at $\Delta V = 0.200$ V. The current is then

$$I = \left(1.00\times10^{-6}\ \text{A}\right)\left(e^{0.200\ \text{V}/0.0250\ \text{V}} - 1\right) = 2.98\times10^{-3}\ \text{A} = \boxed{2.98\ \text{mA}}$$

(d) The ohmic resistance of the diode is

$$R_{\text{ohmic}} = \frac{\Delta V}{I} = \frac{0.200\ \text{V}}{2.98\times10^{-3}\ \text{A}} = \boxed{67.1\ \Omega}$$

(e) The dynamic resistance of the diode is

$$R_{\text{dynamic}} = d(\Delta V)/dI = \left[dI/d(\Delta V)\right]^{-1}$$

where $I = I_0\left(e^{e\Delta V/k_BT} - 1\right)$. Then,

$$\frac{dI}{d(\Delta V)} = \frac{d}{d(\Delta V)}\left[I_0\left(e^{e\Delta V/k_BT} - 1\right)\right] = \frac{eI_0}{k_BT}e^{e\Delta V/k_BT}$$

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Therefore,

$$R_{\text{dynamic}} = \frac{d(\Delta V)}{dI} = \left[\frac{dI}{d(\Delta V)}\right]^{-1} = \left[\frac{eI_0}{k_B T} e^{e\Delta V/k_B T}\right]^{-1}$$

$$= \left[\frac{1.00\times10^{-6}\ \text{A}}{0.0250\ \text{V}} e^{0.200\ \text{V}/0.0250\ \text{V}}\right]^{-1} = \boxed{8.39\ \Omega}$$

P43.49 First, we evaluate I_0 in $I = I_0\left(e^{e\Delta V/k_B T} - 1\right)$, given that I = 200 mA when ΔV = 100 mV and T = 300 K.

$$\frac{e\Delta V}{k_B T} = \frac{\left(1.60\times10^{-19}\ \text{C}\right)(0.100\ \text{V})}{\left(1.38\times10^{-23}\ \text{J/K}\right)(300\ \text{K})} = 3.86$$

so $$I_0 = \frac{I}{e^{e(\Delta V)/k_B T} - 1} = \frac{200\ \text{mA}}{e^{3.86} - 1} = 4.28\ \text{mA}$$

If V = –100 mV, $\frac{e(\Delta V)}{k_B T} = -3.86$; and the current will be

$$I = I_0\left(e^{e\Delta V/k_B T} - 1\right) = (4.28\ \text{mA})\left(e^{-3.86} - 1\right) = \boxed{-4.19\ \text{mA}}$$

P43.50 From Equation 43.27, the current in the diode is a function of ΔV is

$$I(\Delta V) = I_0\left(e^{e\Delta V/k_B T} - 1\right)$$

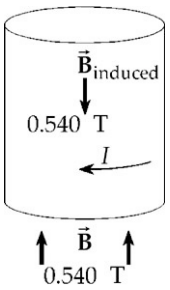
where $k_B T$ = 0.025 0 eV. Therefore,

$$\frac{I(+\Delta V)}{I(-\Delta V)} = \frac{I_0\left(e^{e\Delta V/k_B T} - 1\right)}{I_0\left(e^{e(-\Delta V)/k_B T} - 1\right)} = \frac{e^{e\Delta V/k_B T} - 1}{e^{e(-\Delta V)/k_B T} - 1}$$

$$\frac{I(+1.00\ \text{V})}{I(-1.00\ \text{V})} = \frac{e^{1.00/0.025\,0} - 1}{e^{-1.00/0.025\,0} - 1} = \frac{e^{40} - 1}{e^{-40} - 1} = \boxed{-2.35\times10^{17}}$$

Section 43.8 Superconductivity

P43.51 (a) See ANS. FIG. P43.51.



ANS. FIG. P43.51

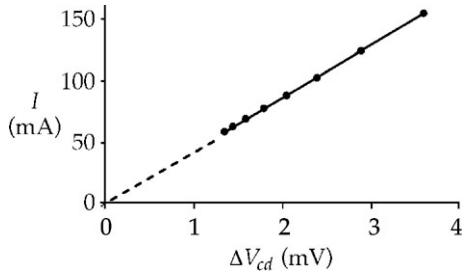
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(b) Treat the rod as a solenoid. For a surface current around the outside of the cylinder as shown, $B = \dfrac{N\mu_0 I}{\ell}$, or

$$NI = \frac{B\ell}{\mu_0} = \frac{(0.540\ \text{T})\left(2.50\times10^{-2}\ \text{m}\right)}{4\pi\times10^{-7}\ \text{T}\cdot\text{m/A}} = \boxed{10.7\ \text{kA}}$$

P43.52 (a) In the definition of resistance, $\Delta V = IR$; if R is zero, then $\Delta V = 0$ for any value of the current.

(b) See ANS. FIG. P43.52. The graph is linear.



ANS. FIG. P43.52

(c) The graph shows a direct proportionality with resistance given by the reciprocal of the slope:

$$\text{slope} = \frac{\Delta I}{\Delta V} = \frac{1}{R} = \frac{(155-57.8)\ \text{mA}}{(3.61-1.356)\ \text{mV}} = 43.1\ \Omega^{-1}$$

so, $R = \boxed{0.023\,2\ \Omega}$

(d) The expulsion of magnetic flux, and therefore fewer current-carrying paths through the superconductor, could explain the decrease in current.

P43.53 By Faraday's law: $\dfrac{\Delta\Phi_B}{\Delta t} = L\dfrac{\Delta I}{\Delta t} = A\dfrac{\Delta B}{\Delta t}$, thus

$$\Delta I = \frac{A(\Delta B)}{L} = \frac{\left[\pi\left(0.010\,0\ \text{m}\right)^2\right]\left(0.020\,0\ \text{T}\right)}{3.10\times10^{-8}\ \text{H}} = 203\ \text{A}$$

The current generated in the ring is 203 A to produce a magnetic field in the direction of the original field through the ring.

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Additional Problems

43.54 For the N_2 molecule, $k = 2\,297$ N/m, $m = 2.32 \times 10^{-26}$ kg, and $r = 1.20 \times 10^{-10}$ m. The reduced mass is, from Equation 43.4,

$$\mu = \frac{mm}{m+m} = \frac{m}{2}$$

The frequency of vibration for the molecule is, from Equation 43.8,

$$\omega = \sqrt{\frac{k}{\mu}} = 4.45 \times 10^{14} \text{ rad/s}$$

and the moment of inertia is, from Equation 43.3,

$$I = \mu r^2 = \left(1.16 \times 10^{-26} \text{ kg}\right)\left(1.20 \times 10^{-10} \text{ m}\right)^2$$
$$= 1.67 \times 10^{-46} \text{ kg} \cdot \text{m}^2$$

The allowed vibrational energies are, from Equation 43.9,

$$E_{\text{vib}} = \left(v + \frac{1}{2}\right)\hbar\omega, \text{ where } v = 1, 2, 3\ldots$$

The first excited vibrational state is above the vibrational ground state by the energy difference $\Delta E = \hbar\omega$. For the rotational state that is above the rotational ground state by the same energy difference, we require

$$\frac{\hbar^2}{2I} J(J+1) = \hbar\omega$$

or

$$J(J+1) = \frac{2I\omega}{\hbar} = \frac{2\left(1.67 \times 10^{-46} \text{ kg} \cdot \text{m}^2\right)\left(4.45 \times 10^{14} \text{ rad/s}\right)}{1.055 \times 10^{-34} \text{ J} \cdot \text{s}}$$
$$= 1\,410.$$

Thus, by inspection, $\boxed{J = 37}$.

P43.55 From Equation 43.9, the allowed vibrational energies are

$$E_{\text{vib}} = \left(v + \frac{1}{2}\right)\hbar\omega, \text{ where } v = 1, 2, 3\ldots$$

For the vibrational energy level that is just below the dissociation energy, we require

$$E_{\text{vib}} = \left(v + \frac{1}{2}\right)\hbar\omega \leq E_{\text{max}} = 4.48 \text{ eV}$$

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or,
$$v \le \frac{E_{\max}}{\hbar\omega} - \frac{1}{2} = \frac{E_{\max}}{\hbar\omega} - \frac{1}{2}$$
$$v \le \frac{(4.48\ \text{eV})\left(1.60\times10^{-19}\ \text{J/eV}\right)}{\left[6.626\times10^{-34}\ \text{J}\cdot\text{s}/2\pi\right]\left(8.28\times10^{14}\ \text{rad/s}\right)} - \frac{1}{2} = 7.7$$

Therefore, because v is an integer, $\boxed{v = 7}$.

P43.56 (a) The total potential energy $U_{\text{total}} = -\alpha\frac{k_e e^2}{r} + \frac{B}{r^m}$, given by Equation 43.17, has its minimum value at the equilibrium spacing, $r = r_0$. At this point, $F = -\left.\frac{dU}{dr}\right|_{r=r_0} = 0$:

$$F = -\frac{d}{dr}\left(-\alpha\frac{k_e e^2}{r} + \frac{B}{r^m}\right)\Bigg|_{r=r_0} = 0$$
$$= -\alpha\frac{k_e e^2}{r_0^2} + \frac{mB}{r_0^{m+1}} = 0$$

which gives

$$B = \alpha\frac{k_e e^2}{m} r_0^{m-1}$$

Substituting this value of B into F, we have

$$F = -\alpha\frac{k_e e^2}{r^2} + \frac{m}{r^{m+1}}\left(\alpha\frac{k_e e^2}{m} r_0^{m-1}\right) = -\alpha\frac{k_e e^2}{r^2}\left[1 - \left(\frac{r_0}{r}\right)^{m-1}\right]$$

(b) Let $r = r_0 + x$, so $r_0 = r - x$. Then assuming x is small we have,

$$F = -\alpha\frac{k_e e^2}{r^2}\left[1 - \left(\frac{r-x}{r}\right)^{m-1}\right] = -\alpha\frac{k_e e^2}{r^2}\left[1 - \left(1 - \frac{x}{r}\right)^{m-1}\right]$$
$$\approx -\alpha\frac{k_e e^2}{r^2}\left[1 - 1 + (m-1)\frac{x}{r}\right] \approx -\alpha\frac{k_e e^2}{r_0^3}(m-1)x$$

This is of the form of Hooke's law with spring constant

$$K = \frac{k_e \alpha e^2}{r_0^3}(m-1).$$

(c) Figure 38.22 (in Section 38.5 on electron diffraction) gives the distance from sodium ion to sodium ion as 0.562 737 nm. Therefore, the interatomic spacing in NaCl is

$$r_0 = (0.562\ 737\ \text{nm})/2 = 0.281\ 369 \times 10^{-9}\ \text{m}$$

Other problems in this chapter give the same information, or we

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could calculate it from the statement in Section 43.3 that the ionic cohesive energy for this crystal is –7.84 eV. Using Equation 43.17,

$$U_0 = -\alpha \frac{k_e e^2}{r_0}\left(1 - \frac{1}{m}\right) = -7.84 \text{ eV}$$

Solving for r_0,

$$r_0 = -\alpha \frac{k_e e^2}{U_0}\left(1 - \frac{1}{m}\right)$$

$$= -\alpha \frac{\left(8.99 \times 10^9 \text{ N} \cdot \text{m}^2/\text{C}^2\right)\left(1.60 \times 10^{-19} \text{ C}\right)^2}{\left(-7.84 \text{ eV}\right)\left(1.60 \times 10^{-19} \text{ J/eV}\right)}\left(1 - \frac{1}{8}\right)$$

$$= 2.81 \times 10^{-10} \text{ m}$$

The stiffness constant is then

$$K = \alpha \frac{k_e e^2}{r_0^3}(m-1)$$

$$= \left(1.747\,6\right)\frac{\left(8.99 \times 10^9 \text{ N} \cdot \text{m}^2/\text{C}^2\right)(1.60 \times 10^{-19} \text{ C})^2(8-1)}{(2.81 \times 10^{-10} \text{ m})^3}$$

$$= 127 \text{ N/m}$$

The vibration frequency of a sodium ion (m = 23.0 u) within the crystal is

$$f = \frac{1}{2\pi}\sqrt{\frac{K}{m}} = \frac{1}{2\pi}\sqrt{\frac{127 \text{ N/m}}{23.0\left(1.66 \times 10^{-27} \text{ kg}\right)}}$$

$$= 9.18 \times 10^{12} \text{ Hz} = \boxed{9.18 \text{ THz}}$$

P43.57 Because the average energy required to break one van der Waals bond is 1.74×10^{-23} J, and because the bond is between two atoms of the same kind, the energy required to remove one helium atom from the bond is half the total:

$$1.74 \times 10^{-23} \text{ J}/2 = 0.870 \times 10^{-23} \text{ J}$$

Because each atom bonds with four other atoms, the energy required to remove one atom from all four bonds is

$$4\left(0.870 \times 10^{-23} \text{ J}\right) = 3.48 \times 10^{-23} \text{ J/atom}$$

The latent heat of fusion for helium (in joules per gram) is the total energy required to break the bonds of all the helium atoms in a mol, expressed as energy/ unit mass:

$$L = \left(\frac{3.48 \times 10^{-23} \text{ J}}{\text{atom}}\right)\left(\frac{6.02 \times 10^{23} \text{ atoms}}{\text{mol}}\right)\left(\frac{1 \text{ mol}}{4.00 \text{ g}}\right) = \boxed{5.24 \text{ J/g}}$$

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P43.58 We assume the potential well is that of a harmonic-oscillator. From Equation 43.9, the allowed energies of vibration of the molecule are

$$E_{\text{vib}} = \left(v + \frac{1}{2} \right) hf$$

To dissociate the atoms, enough energy must be supplied to raise their energy to the top of the potential well. The energy required to dissociate the atoms in the ground state ($v = 0$) is 4.48 eV; thus, the well depth is $\frac{1}{2}hf + 4.48 \text{ eV}$. In the first excited vibrational state ($v = 1$), the dissociation energy is 3.96 eV; thus, the well depth is $\frac{3}{2}hf + 3.96 \text{ eV}$. Then, the depth of the well is

$$\frac{1}{2}hf + 4.48 \text{ eV} = \frac{3}{2}hf + 3.96 \text{ eV}$$

from which we see that $hf = 0.52$ eV. Therefore, the depth of the well is

$$\frac{1}{2}hf + 4.48 \text{ eV} = \frac{1}{2}(0.520 \text{ eV}) + 4.48 \text{ eV} = \boxed{4.74 \text{ eV}}$$

P43.59 The total potential energy is given by Equation 43.17:

$$U_{\text{total}} = -\alpha \frac{k_e e^2}{r} + \frac{B}{r^m}$$

The total potential energy has its minimum value U_0 at the equilibrium spacing, $r = r_0$. At this point, $\left. \frac{dU}{dr} \right|_{r=r_0} = 0,$ or

$$\left. \frac{dU}{dr} \right|_{r=r_0} = \frac{d}{dr}\left(-\alpha \frac{k_e e^2}{r} + \frac{B}{r^m} \right)\Bigg|_{r=r_0} = -\alpha \frac{k_e e^2}{r_0^2} + \frac{mB}{r_0^{m+1}} = 0$$

which gives

$$B = \alpha \frac{k_e e^2}{m} r_0^{m-1}$$

Substituting this value of B into U_{total}, we arrive at

$$U_0 = -\alpha \frac{k_e e^2}{r_0} + \alpha \frac{k_e e^2}{m} r_0^{m-1}\left(\frac{1}{r_0^m} \right) = -\alpha \frac{k_e e^2}{r_0}\left(1 - \frac{1}{m} \right)$$

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P43.60 (a) The results of the spreadsheet are shown in two parts, in TABLE P43.60(a) and TABLE P43.60(b) for $f(E) = \dfrac{1}{e^{\left[(E/E_F - 1)T_F\right]/T} + 1}$. ANS. FIG. P43.60 shows the graphs of the tabulated values.

(b) The function is compared to the case $T = 0$. See the table and graphs below.

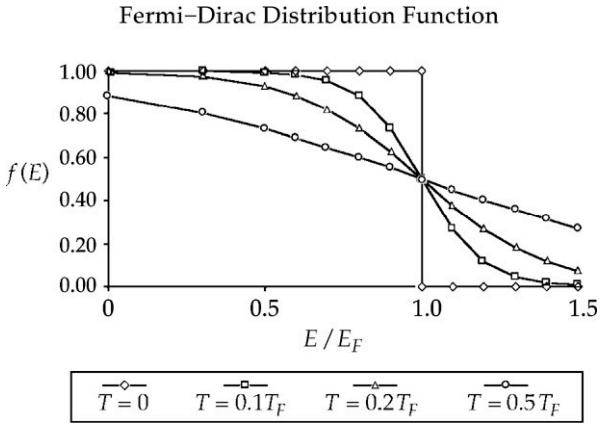
	$T = 0$		$T = 0.1T_F$	
$\dfrac{E}{E_F}$	$e^{\left[(E/E_F)-1\right](T_F/T)}$	$f(E)$	$e^{\left[(E/E_F)-1\right](T_F/T)}$	$f(E)$
0	$e^{-\infty}$	1.00	$e^{-10.0}$	1.000
0.500	$e^{-\infty}$	1.00	$e^{-5.00}$	0.993
0.600	$e^{-\infty}$	1.00	$e^{-4.00}$	0.982
0.700	$e^{-\infty}$	1.00	$e^{-3.00}$	0.953
0.800	$e^{-\infty}$	1.00	$e^{-2.00}$	0.881
0.900	$e^{-\infty}$	1.00	$e^{-1.00}$	0.731
1.00	e^{0}	0.500	e^{0}	0.500
1.10	$e^{+\infty}$	0.00	$e^{1.00}$	0.269
1.20	$e^{+\infty}$	0.00	$e^{2.00}$	0.119
1.30	$e^{+\infty}$	0.00	$e^{3.00}$	0.047 4
1.40	$e^{+\infty}$	0.00	$e^{4.00}$	0.018 0
1.50	$e^{+\infty}$	0.00	$e^{5.00}$	0.006 69

TABLE P43.60(a)

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	$T = 0.2T_F$		$T = 0.5T_F$	
$\frac{E}{E_F}$	$e^{[(E/E_F)-1](T_F/T)}$	$f(E)$	$e^{[(E/E_F)-1](T_F/T)}$	$f(E)$
0	$e^{-5.00}$	0.993	$e^{-2.00}$	0.881
0.500	$e^{-2.50}$	0.924	$e^{-1.00}$	0.731
0.600	$e^{-2.00}$	0.881	$e^{-0.800}$	0.690
0.700	$e^{-1.50}$	0.818	$e^{-0.600}$	0.646
0.800	$e^{-1.00}$	0.731	$e^{-0.400}$	0.599
0.900	$e^{-0.500}$	0.622	$e^{-0.200}$	0.550
1.00	e^{0}	0.500	e^{0}	0.500
1.10	$e^{0.500}$	0.378	$e^{0.200}$	0.450
1.20	$e^{1.00}$	0.269	$e^{0.400}$	0.401
1.30	$e^{1.50}$	0.182	$e^{0.600}$	0.354
1.40	$e^{2.00}$	0.119	$e^{0.800}$	0.310
1.50	$e^{2.50}$	0.075 9	$e^{1.00}$	0.269

TABLE P43.60(b)



ANS. FIG. P43.60

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P43.61 (a) For equilibrium, $\frac{dU}{dx} = 0$:

$$\frac{d}{dx}\left(Ax^{-3} - Bx^{-1}\right) = -3Ax^{-4} + Bx^{-2} = 0$$

$x \to \infty$ describes one equilibrium position, but the stable equilibrium position x_0 is at

$$-3Ax_0^{-4} + Bx_0^{-2} = 0$$

solving,

$$x_0^2 = \frac{3A}{B} \quad \rightarrow \quad x_0 = \sqrt{\frac{3A}{B}} = \sqrt{\frac{3\left(0.150\ \text{eV}\cdot\text{nm}^3\right)}{3.68\ \text{eV}\cdot\text{nm}}} = \boxed{0.350\ \text{nm}}$$

(b) The depth of the well is given by

$$U_0 = U\big|_{x=x_0} = \frac{A}{x_0^3} - \frac{B}{x_0} = \frac{AB^{3/2}}{3^{3/2}A^{3/2}} - \frac{BB^{1/2}}{3^{1/2}A^{1/2}}$$

$$U_0 = U\big|_{x=x_0} = -\frac{2B^{3/2}}{3^{3/2}A^{1/2}} = -\frac{2\left(3.68\ \text{eV}\cdot\text{nm}\right)^{3/2}}{3^{3/2}\left(0.150\ \text{eV}\cdot\text{nm}^3\right)^{1/2}} = \boxed{-7.02\ \text{eV}}$$

(c) The force on the particle is given by $F_x = -\frac{dU}{dx} = 3Ax^{-4} - Bx^{-2}$. To find the maximum force, we determine finite x_m such that $\left.\frac{dF}{dx}\right|_{x=x_m} = 0$.

$$\left.\frac{dF_x}{dx}\right|_{x=x_m} = \left[-12Ax^{-5} + 2Bx^{-3}\right]_{x=x_0} = 0$$

$$2Bx_m^{-3} = 12Ax_m^{-5}$$

$$x_m^2 = \frac{6A}{B} \qquad \rightarrow \qquad x_m = \sqrt{\frac{6A}{B}}$$

Then,

$$F_{\max} = 3A\left(\frac{B}{6A}\right)^2 - B\left(\frac{B}{6A}\right) = -\frac{B^2}{12A} = -\frac{\left(3.68\ \text{eV}\cdot\text{nm}\right)^2}{12\left(0.150\ \text{eV}\cdot\text{nm}^3\right)}$$

so $$F_{\max} = -7.52\ \text{eV/nm}\left(\frac{1.60\times10^{-19}\ \text{J}}{1\ \text{eV}}\right)\left(\frac{1\ \text{nm}}{10^{-9}\ \text{m}}\right) = -1.20\times10^{-9}\ \text{N}$$

$$= -1.20\ \text{nN}$$

or, as a vector, $\boxed{-1.20\hat{\mathbf{i}}\ \text{nN}}$.

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P43.62 (a) For equilibrium, $\frac{dU}{dx} = 0$:

$$\frac{d}{dx}(Ax^{-3} - Bx^{-1}) = -3Ax^{-4} + Bx^{-2} = 0$$

$x \to \infty$ describes one equilibrium position, but the stable equilibrium position x_0 is at

$$-3Ax_0^{-4} + Bx_0^{-2} = 0$$

$$Bx_0^{-2} = 3Ax_0^{-4}$$

$$x_0^2 = \frac{3A}{B} \quad \rightarrow \quad \boxed{x_0 = \sqrt{\frac{3A}{B}}}$$

(b) The depth of the well is given by

$$U_0 = U|_{x=x_0} = \frac{A}{x_0^3} - \frac{B}{x_0} = \frac{AB^{3/2}}{3^{3/2}A^{3/2}} - \frac{BB^{1/2}}{3^{1/2}A^{1/2}} = \boxed{-2\sqrt{\frac{B^3}{27A}}}$$

(c) The force on the particle is given by $F_x = -\frac{dU}{dx} = 3Ax^{-4} - Bx^{-2}$.

To find the maximum force, we determine finite x_m such that

$$\left.\frac{dF_x}{dx}\right|_{x=x_m} = \left[-12Ax^{-5} + 2Bx^{-3}\right]_{x=x_0} = 0$$

$$2Bx_m^{-3} = 12Ax_m^{-5}$$

$$x_m^2 = \frac{6A}{B} \quad \rightarrow \quad x_m = \sqrt{\frac{6A}{B}}$$

then

$$F_{\max} = 3A\left(\frac{B}{6A}\right)^2 - B\left(\frac{B}{6A}\right) = \boxed{-\frac{B^2}{12A}}$$

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Challenge Problems

P43.63 (a) Refer to Example 43.2 for details. Since the interatomic potential is the same for both molecules, the spring constant is the same.

Then,

$$f = \frac{1}{2\pi}\sqrt{\frac{k}{\mu}}$$

where

$$\mu_{12} = \frac{(12\text{ u})(16\text{ u})}{12\text{ u} + 16\text{ u}} = 6.86\text{ u} \quad \text{and} \quad \mu_{14} = \frac{(14\text{ u})(16\text{ u})}{14\text{ u} + 16\text{ u}} = 7.47\text{ u}$$

Therefore,

$$f_{14} = \frac{1}{2\pi}\sqrt{\frac{k}{\mu_{14}}} = \frac{1}{2\pi}\sqrt{\frac{k}{\mu_{12}}\left(\frac{\mu_{12}}{\mu_{14}}\right)} = f_{12}\sqrt{\frac{\mu_{12}}{\mu_{14}}}$$

$$= \left(6.42\times10^{13}\text{ Hz}\right)\sqrt{\frac{6.86\text{ u}}{7.47\text{ u}}}$$

$$= \boxed{6.15\times10^{13}\text{ Hz}}$$

(b) The equilibrium distance is the same for both molecules.

$$I_{14} = \mu_{14}r^2 = \left(\frac{\mu_{14}}{\mu_{12}}\right)\mu_{12}r^2 = \left(\frac{\mu_{14}}{\mu_{12}}\right)I_{12}$$

$$I_{14} = \left(\frac{7.47\text{ u}}{6.86\text{ u}}\right)\left(1.46\times10^{-46}\text{ kg}\cdot\text{m}^2\right) = \boxed{1.59\times10^{-46}\text{ kg}\cdot\text{m}^2}$$

(c) The molecule can move to the ($v = 1$, $J = 9$) state or to the ($v = 1$, $J = 11$) state. The energy it can absorb is either

$$\Delta E = \frac{hc}{\lambda} = \left[\left(1+\frac{1}{2}\right)hf_{14} + 11(11+1)\frac{\hbar^2}{2I_{14}}\right] - \left[\left(0+\frac{1}{2}\right)hf_{14} + 10(10+1)\frac{\hbar^2}{2I_{14}}\right]$$

$$\frac{hc}{\lambda} = hf_{14} + 22\frac{\hbar^2}{2I_{14}} = hf_{14} + 11\frac{h\hbar}{2\pi I_{14}}$$

$$\frac{c}{\lambda} = f_{14} + 11\frac{\hbar}{2\pi I_{14}}$$

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or

$$\Delta E = \frac{hc}{\lambda} = \left[\left(1+\frac{1}{2}\right)hf_{14} + 9(9+1)\frac{\hbar^2}{2I_{14}}\right] - \left[\left(0-\frac{1}{2}\right)hf_{14} + 10(10+1)\frac{\hbar^2}{2I_{14}}\right]$$

$$\frac{hc}{\lambda} = hf_{14} - 20\frac{\hbar^2}{2I_{14}} = hf_{14} - 10\frac{h\hbar}{2\pi I_{14}}$$

$$\frac{c}{\lambda} = f_{14} - 10\frac{\hbar}{2\pi I_{14}}$$

The wavelengths it can absorb are then

$$\lambda = \frac{c}{f_{14} + 11\hbar/(2\pi I_{14})} \quad \text{or} \quad \lambda = \frac{c}{f_{14} - 10\hbar/(2\pi I_{14})}$$

These are,

$$\lambda = \frac{2.998\times10^{8}\ \text{m/s}}{6.15\times10^{13}\ \text{Hz} + \dfrac{\left[11\left(1.055\times10^{-34}\ \text{J}\cdot\text{s}\right)\right]}{\left[2\pi\left(1.59\times10^{-46}\ \text{kg}\cdot\text{m}^2\right)\right]}} = \boxed{4.78\ \mu\text{m}}$$

or

$$\lambda = \frac{2.998\times10^{8}\ \text{m/s}}{6.15\times10^{13}\ \text{Hz} - \dfrac{\left[10\left(1.055\times10^{-34}\ \text{J}\cdot\text{s}\right)\right]}{\left[2\pi\left(1.59\times10^{-46}\ \text{kg}\cdot\text{m}^2\right)\right]}} = \boxed{4.96\ \mu\text{m}}$$

P43.64 (a) At equilibrium separation $r = r_e$,

$$\left.\frac{dU}{dr}\right|_{r=r_e} = -2aB\left[e^{-a(r_e-r_0)} - 1\right]e^{-a(r_e-r_0)} = 0$$

We have neutral equilibrium as $r_e \to \infty$ and stable equilibrium at

$$e^{-a(r_e-r_0)} = 1 \qquad \rightarrow \qquad r_e = \boxed{r_0}$$

(b) At $r = r_0$, $U = 0$. As $r \to \infty$, $U \to B$. The depth of the well is $\boxed{B}$.

(c) We expand the potential in a Taylor series about the equilibrium point $r = r_0$:

$$U(r) \approx U(r_0) + \left.\frac{dU}{dr}\right|_{r=r_0}(r-r_0) + \frac{1}{2}\left.\frac{d^2U}{dr^2}\right|_{r=r_0}(r-r_0)^2$$

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or,

$$U(r) \approx 0 + 0 + \frac{1}{2}(-2Ba)\left[-2ae^{-2a(r-r_0)} + ae^{-a(r-r_0)}\right]_{r=r_0}(r-r_0)^2$$
$$\approx Ba^2(r-r_0)^2$$

This is of the form

$$\frac{1}{2}kx^2 = \frac{1}{2}k(r-r_0)^2$$

for a simple harmonic oscillator with $k = 2Ba^2$.

Then, the molecule vibrates with frequency

$$f = \frac{1}{2\pi}\sqrt{\frac{k}{\mu}} = \frac{a}{2\pi}\sqrt{\frac{2B}{\mu}} = \boxed{\frac{a}{\pi}\sqrt{\frac{B}{2\mu}}}$$

(d) The ground state energy is

$$\frac{1}{2}\hbar\omega = \frac{1}{2}hf = \frac{ha}{\pi}\sqrt{\frac{B}{8\mu}}$$

The energy at infinity is B. Therefore, to separate the nuclei to infinity requires energy

$$\boxed{B - \frac{ha}{\pi}\sqrt{\frac{B}{8\mu}}}$$

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ANSWERS TO EVEN-NUMBERED PROBLEMS

P43.2 (a) 921 pN toward the other ion; (b) −2.88 eV

P43.4 (a) $E_a = 1.28$ eV; (b) $\sigma = 0.272$ nm, $\epsilon = 4.65$ eV; (c) +6.55 nN; (d) 576 N/m

P43.6 (a) $r_0 = \left[\frac{2A}{B}\right]^{1/6}$; (b) $\frac{B^2}{4A}$

P43.8 (a) 0.120 meV; (b) 19.3 GHz

P43.10 (a) 1.22×10^{-26} kg; (b) 1.24×10^{-26} kg; (c) They agree because the small apparent difference can be attributed to uncertainty in the data.

P43.12 The incident photons have a wavelength longer than this, which means they have less energy than 0.359 eV. Therefore, these photons cannot excite the molecule to the first excited state.

P43.14 $2.72 \times 10^{-47}\ \text{kg} \cdot \text{m}^2$

P43.16 μr^2

P43.18 (a) $1.89 \times 10^{-45}\ \text{kg} \cdot \text{m}^2$; (b) $E_{\text{rot}} = 18.4\, J(J+1)$, where E_{rot} is in microelectron volts and $J = 0, 1, 2, 3, \ldots$.

P43.20 $2.88 \times 10^{-47}\ \text{kg} \cdot \text{m}^2$

P43.22 64.1 THz

P43.24 (a) $\sim 10^{17}$; (b) $\sim 10^5\ \text{m}^3$

P43.26 $U = -k_e \alpha \frac{e^2}{r}$ where $\alpha = 2 \ln 2$

P43.28 (a) The Fermi energy is proportional to the spatial concentration of free electrons to the two-thirds power; (b) See P43.28(b) for full explanation; (c) 6.04; (d) Copper; (e) 0.333; (f) This behavior agrees with the proportionality because $E_{\text{F}} \sim n_e^{2/3}$ and $6.04^{2/3} = 3.32$.

P43.30 (a) 1.57 Mm/s; (b) The speed is larger by ten orders of magnitude.

P43.32 3.40×10^{17} electrons

P43.34 There are approximately two free electrons per atom for this metal, not one (see P43.34 for full explanation).

P43.36 $\frac{1}{e^{(\beta - 1)E_F / k_B T} + 1}$

P43.38 (a) 1.10; (b) 9.42×10^{-25}

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P43.40 (a) The gap should be less than or equal to 1.24 eV; (b) Because silicon has an energy gap of 1.14 eV, it can absorb the energy of nearly all of the photons in sunlight and is an appropriate material for a solar energy collector.

P43.42 (a) All the hydrogen Balmer lines except for the red lien at 656 nm will be absorbed; (b) The red line at 656 nm will be transmitted.

P43.44 2.42 eV

P43.46 (a) $a' = \left(\frac{m_e}{m^*}\right)\kappa a_0$; (b) 2.81 nm; (c) $E_n' = -\left(\frac{m^*}{m_e}\right)\frac{E_n}{\kappa^2}$; (d) −0.0219 eV

P43.48 (a) See P43.48(a) for full explanation; (b) See ANS. FIG. P43.48; (c) 2.98 mA; (d) 67.1 Ω; (e) 8.39 Ω

P43.50 -2.35×10^{17}

P43.52 (a) In the definition of resistance $\Delta V = IR$, if R is zero then $\Delta V = 0$ for any value of current; (b) See ANS FIG P43.52; (c) 0.023 2 Ω; (d) Expulsion of magnetic flux, and therefore fewer current-carrying paths through the superconductor, could explain the decrease in current.

P43.54 $J = 37$

P43.56 (a) See P43.56(a) for full explanation; (b) See P43.56(b) for full explanation; (c) 9.18 THz

P43.58 4.74 eV

P43.60 (a–b) See P43.60 for full explanation.

P43.62 (a) $x_0 = \sqrt{\frac{3A}{B}}$; (b) $-2\sqrt{\frac{B^3}{27A}}$; (c) $-\frac{B^2}{12A}$

P43.64 (a) r_0; (b) B; (c) $\frac{a}{\pi}\sqrt{\frac{B}{2\mu}}$; (d) $B - \frac{ha}{\pi}\sqrt{\frac{B}{8\mu}}$

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